

NSEJS CHEMISTRY

CHEMICAL BONDING

ONE SHOT



Nidhi Prashar

Chemistry



- **16+ Years of teaching experience**
- **Gold Medallist in the Master's Degree (Chemistry), UGC NET Qualified**
- **IJSO EXPERT with 170+ NSEJS Selections**
- **Mentored Science Olympiads GOLD MEDALISTS**
- **Teaching and mentoring students IIT, NEET, NSEC, IJSO exams**
- **Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)**

Vedantu

Vedantu Creates History in

International Olympiads 2020-26

2

Gold Medals

IPho

(International
Physics Olympiad)

2

Gold Medals

IBO

(International
Biology Olympiad)

1

Gold Medal

IOAA

(International Olympiad
on Astronomy & Astrophysics)

1

Gold Medal

IOAA (Jr)

(International Olympiad
on Astronomy & Astrophysics Jr)

1

Gold Medal

APho

(Asian Physics Olympiad)

1

Gold Medals

IJSO

(International Junior
Science Olympiad)

2

Silver Medals

IOAA

(International Olympiad
on Astronomy & Astrophysics)

2

Silver Medals

ICho

(International
Chemistry Olympiad)

1

Silver Medal

IJSO

(International Junior
Science Olympiad)

1

Silver Medal

APho

(Asian
Physics Olympiad)

1

Bronze Medal

IOAA (Jr)

(International Olympiad
on Astronomy & Astrophysics Jr)



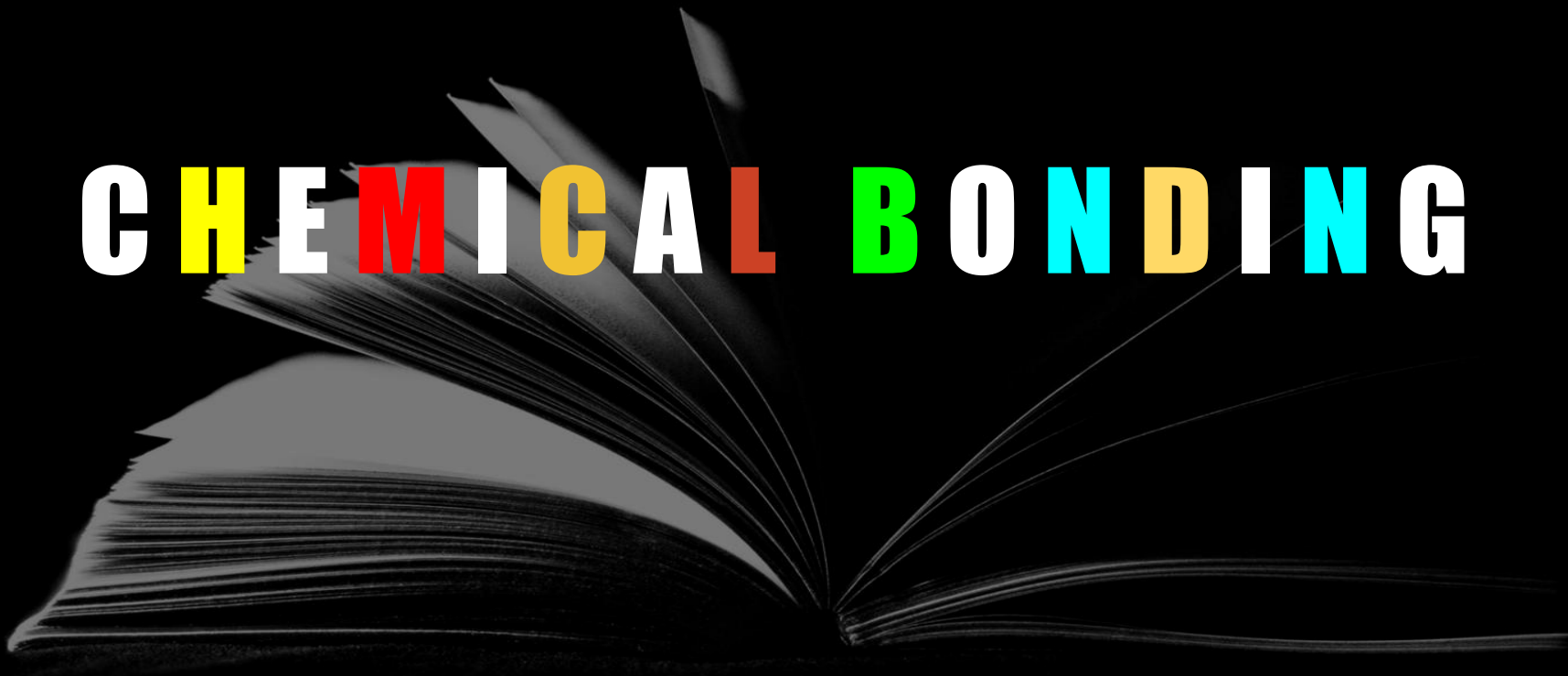
Vedantu Today. Olympiad Champions Tomorrow

What we will be covering in the session

- *Complete Chapter theory*
- *All Important concepts*
- *NSEJS / SEHSS PYQ's*
- *Important Questions*

Telegram Channel: https://t.me/Vedantu_Science_Olympiad

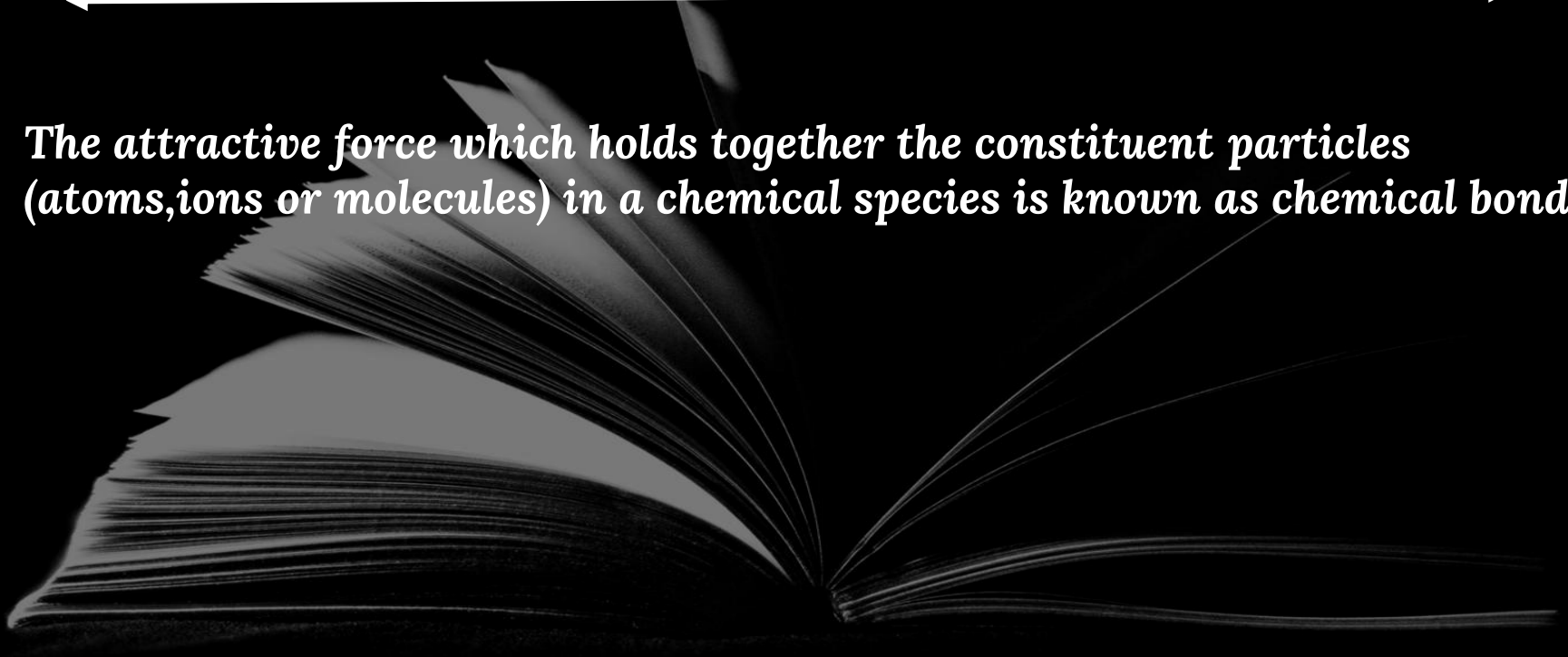
CHEMICAL BONDING



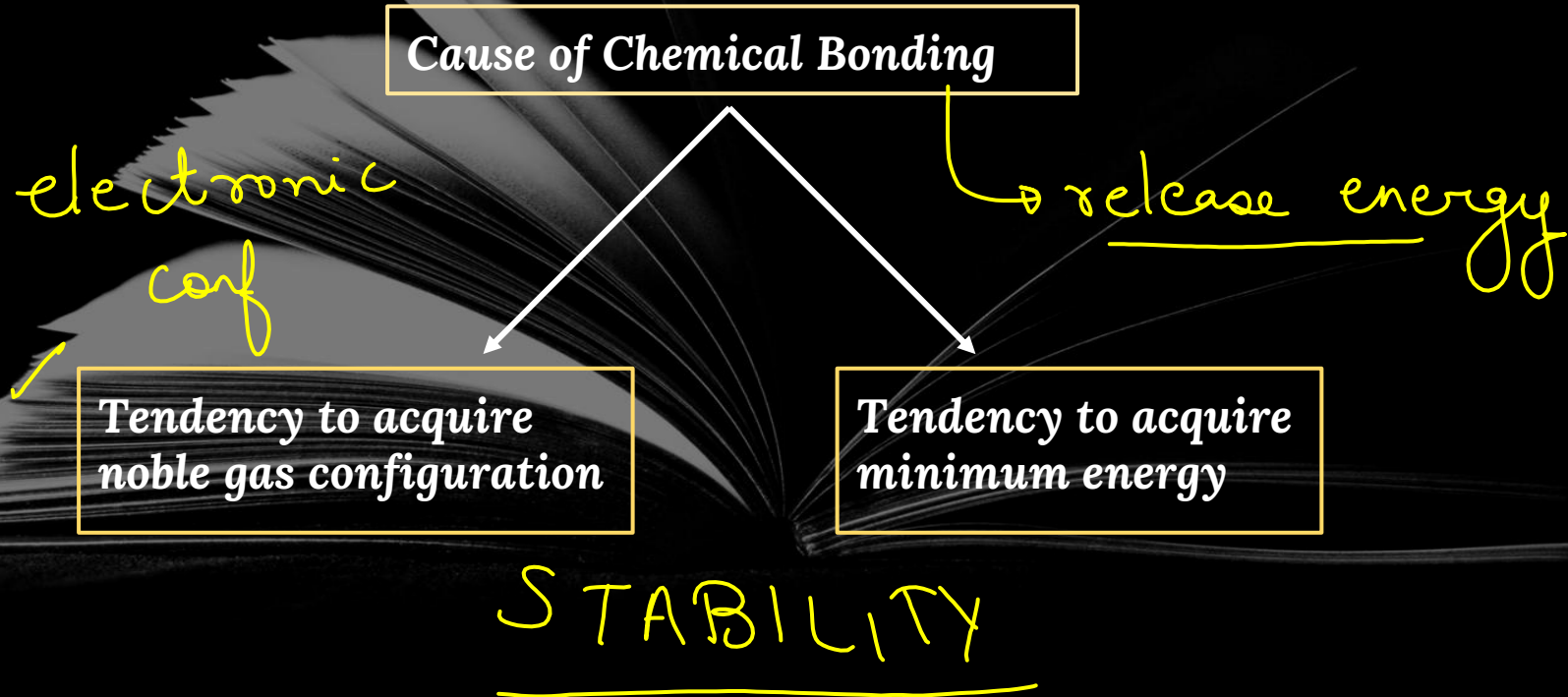
CHEMICAL BOND



The attractive force which holds together the constituent particles (atoms, ions or molecules) in a chemical species is known as chemical bond.



CHEMICAL BOND



Chemical bonding

Strong bond

Weak bond

Ionic
bond

Covalent
bond

Coordinate
bond

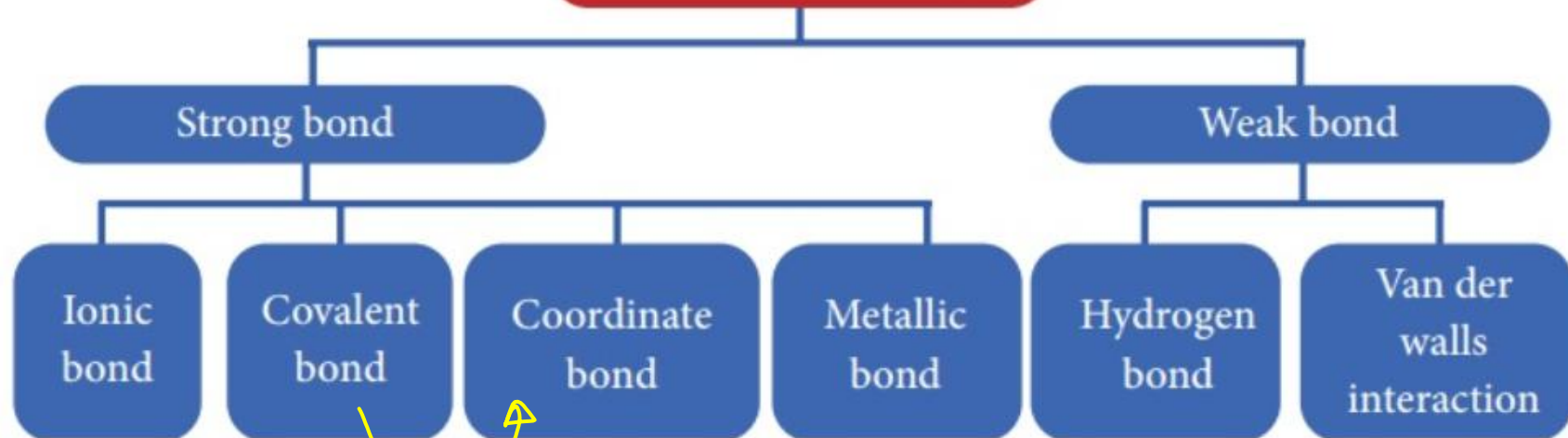
Metallic
bond

Hydrogen
bond

Van der
walls
interaction

Inter atomic

Inter molecule



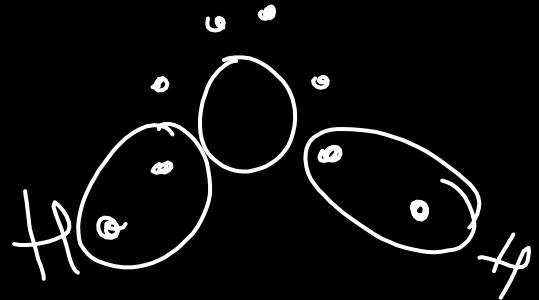
LEWIS OCTET RULE

- Every atom has a tendency to complete its octet in outer most shell.
- H has the tendency to complete its duplet.
- To acquire inert gas configuration atoms loose or gain electron or share electron



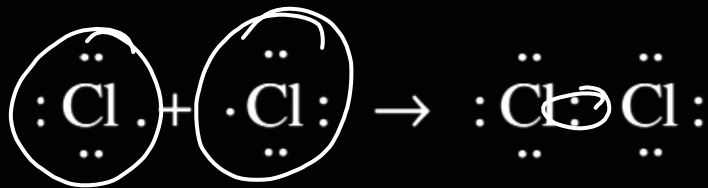
LEWIS OCTET RULE

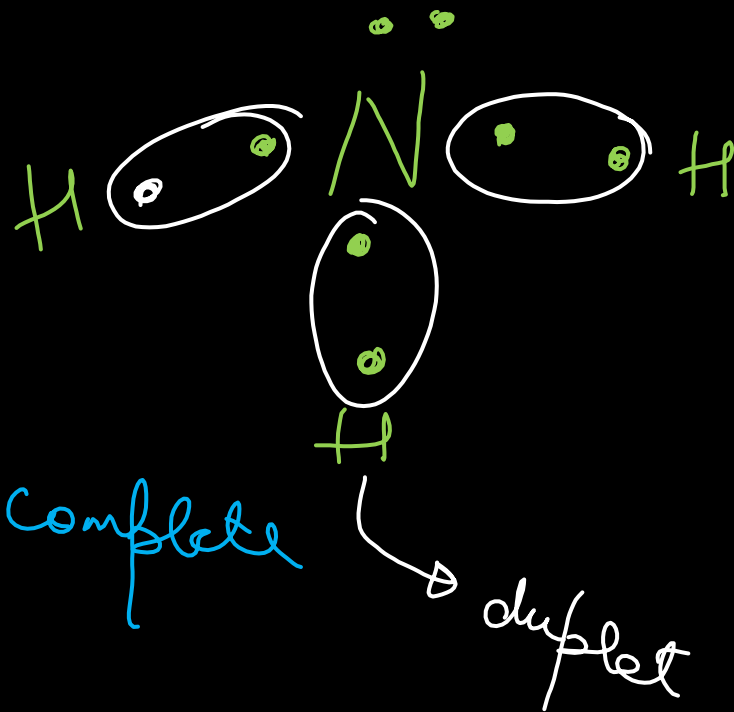
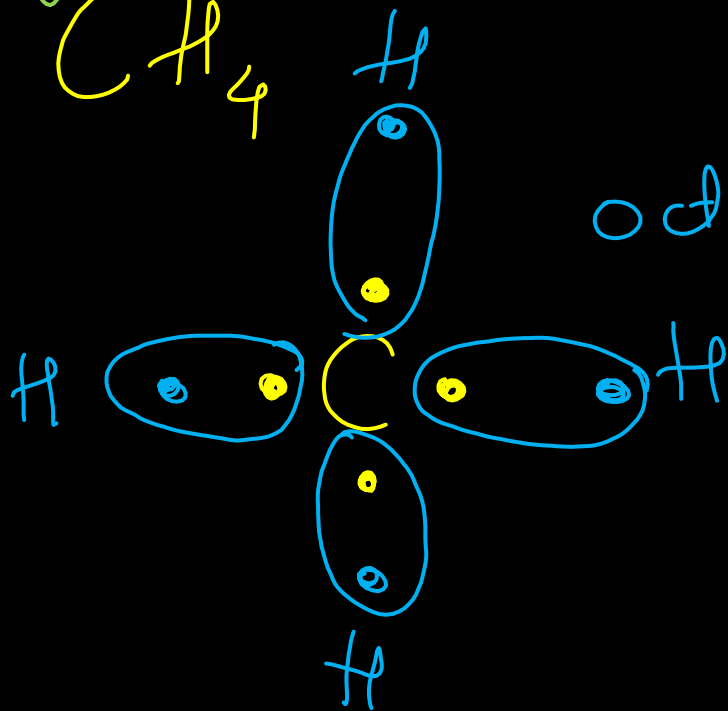
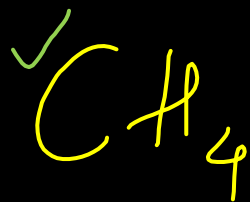
The tendency of atoms to achieve eight electrons in their outermost shell is known as **Lewis octet rule**



LEWIS OCTET RULE

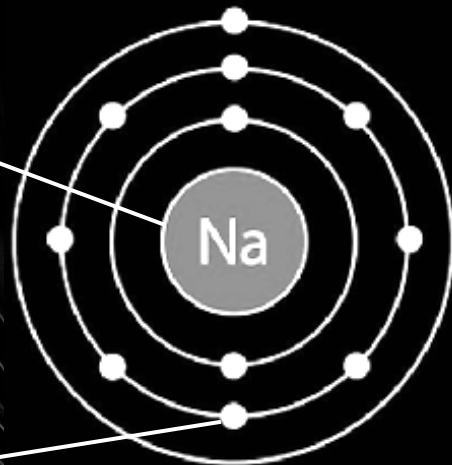
A chlorine atom has seven electrons in its outer shell, so by sharing one electron with another chlorine atom both atoms attain an octet and form a chlorine molecule Cl_2 .



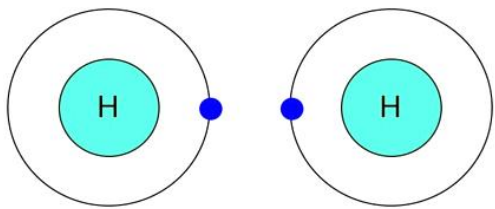


LEWIS OCTET RULE

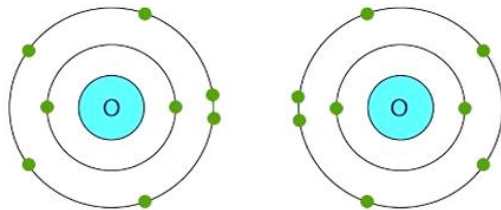
Positive Kernal
(Nucleus)



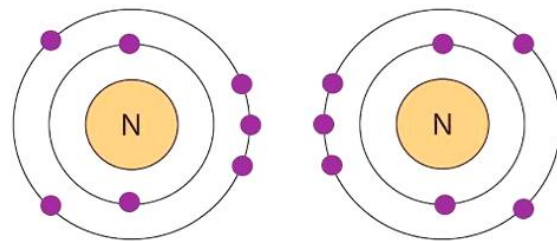
Dots
(Representing e's)



Single bond
 H_2



Double bond
 O_2



Triple bond
 N_2

LEWIS DOT STRUCTURE



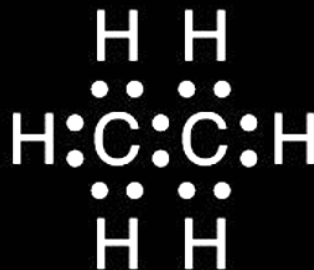
Ethane (C_2H_6)

Valence
electrons

$$\text{C} = 4 \times 2 = 8$$

$$\text{H} = 1 \times 6 = 6$$

$$\text{Total} = 14 \text{ e}^-$$



LEWIS DOT STRUCTURE



$$\text{N} = 1 \times 5 = 5$$

$$\text{O} = 2 \times 6 = 12$$

$$\text{Total} = 17 \text{ e}^-$$

Valence
electrons



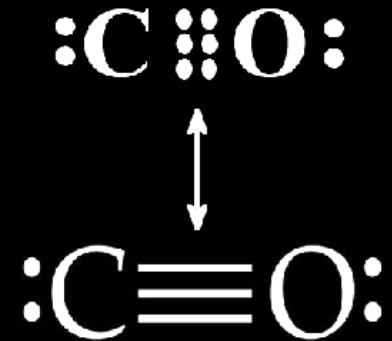
Lewis dot structures



CO

Valence
electrons

$$\begin{array}{r} \text{C} = 1 \times 4 = 4 \\ \text{O} = 1 \times 6 = 6 \\ \hline \text{Total} = 10 \text{ e}^- \\ \hline \end{array}$$



Molecule/Ion		Lewis Representation
H_2	$\text{H} : \text{H}^*$	$\text{H} - \text{H}$
O_2	$:\ddot{\text{O}}::\ddot{\text{O}}:$	$:\ddot{\text{O}}=\ddot{\text{O}}:$
O_3	$ \begin{array}{c} \ddot{\text{O}}^+ \\ \diagup \quad \diagdown \\ \ddot{\text{O}} \quad \ddot{\text{O}}^- \end{array} $	$ \begin{array}{c} \ddot{\text{O}}^+ \\ \diagup \quad \diagdown \\ \ddot{\text{O}} \quad \ddot{\text{O}}^- \end{array} $
NF_3	$ \begin{array}{c} :\ddot{\text{F}}: \quad :\ddot{\text{N}}: \quad :\ddot{\text{F}}: \\ \\ :\ddot{\text{F}}: \end{array} $	$ \begin{array}{c} :\ddot{\text{F}}- \ddot{\text{N}} - \ddot{\text{F}}: \\ \\ :\ddot{\text{F}}: \end{array} $
CO_3^{2-}	$ \left[\begin{array}{ccc} & :\ddot{\text{O}}: & \\ :\ddot{\text{O}}: & \text{C} & :\ddot{\text{O}}: \\ & & \end{array} \right]^{2-} $	$ \left[\begin{array}{ccc} & :\ddot{\text{O}}: & \\ :\ddot{\text{O}}- & \text{C} & -\ddot{\text{O}}: \\ & & \end{array} \right]^{2-} $
HNO_3	$ \begin{array}{c} \ddot{\text{O}}: \quad \text{N}^+ \quad \ddot{\text{O}}: \text{H} \\ \\ :\ddot{\text{O}}^-: \end{array} $	$ \begin{array}{c} \ddot{\text{O}} = \text{N}^+ - \ddot{\text{O}} - \text{H} \\ \\ :\ddot{\text{O}}^-: \end{array} $

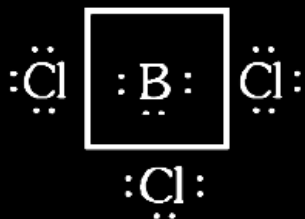
Limitations of Octet Rule



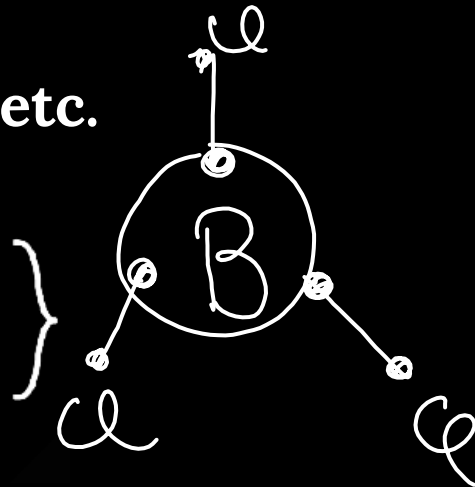
Incomplete octet molecules :- or (electron deficient molecules)

Compound in which octet is not complete in outer most orbit of central atom.

Example - Halides of IIIA groups, BF_3 , AlCl_3 , BCl_3 etc.



{ In BCl_3
Boron has only 6 electrons }



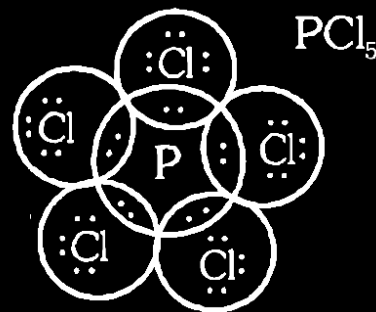
Limitations of Octet Rule

Expansion of octet or (electron efficient molecules)

Compound in which central atom has more than $8e^-$ in outermost orbits.

Example - In PCl_5 , SF_6 and IF_7 the central atom P, S and I contain 10, 12, and 14 electrons respectively.

expanded octet/
super octet



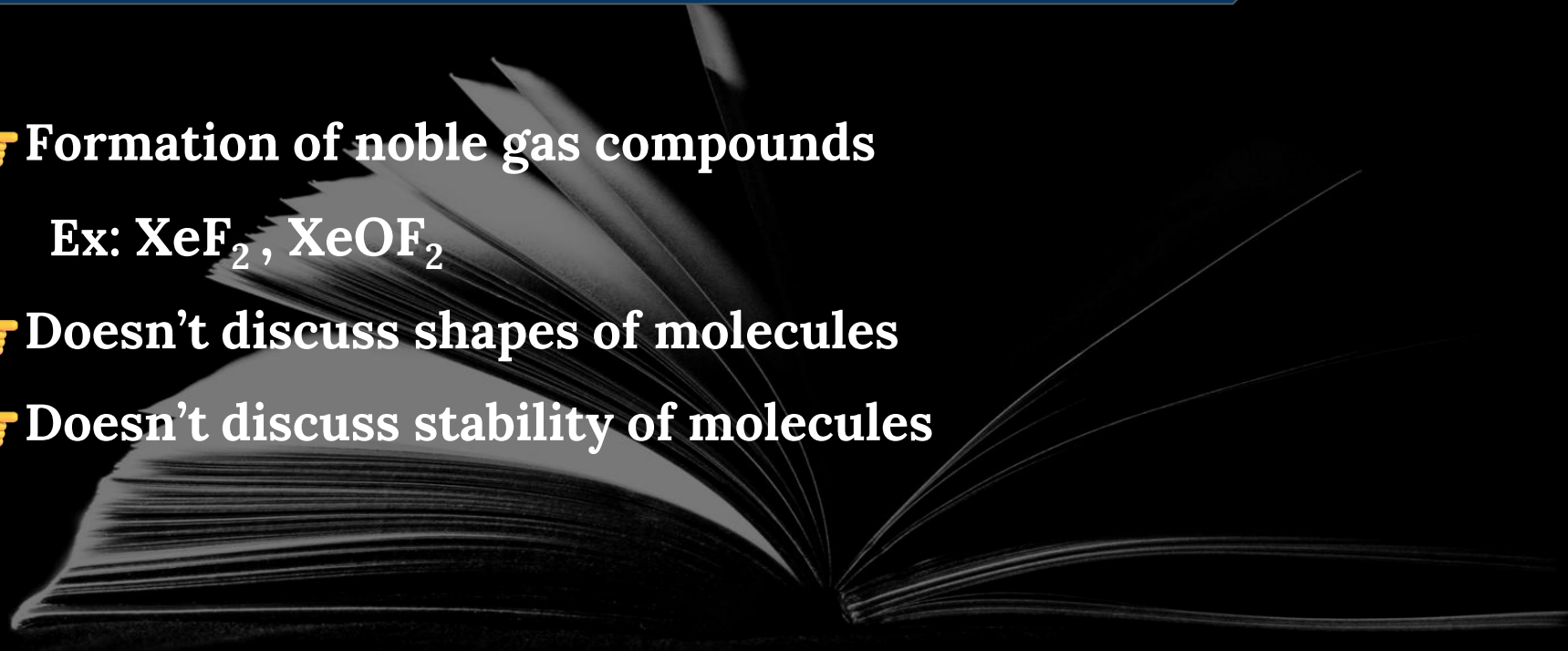
Limitations of Octet Rule

👉 Formation of noble gas compounds

Ex: XeF_2 , XeOF_2

👉 Doesn't discuss shapes of molecules

👉 Doesn't discuss stability of molecules



Which is not an exception to octet rule?



A



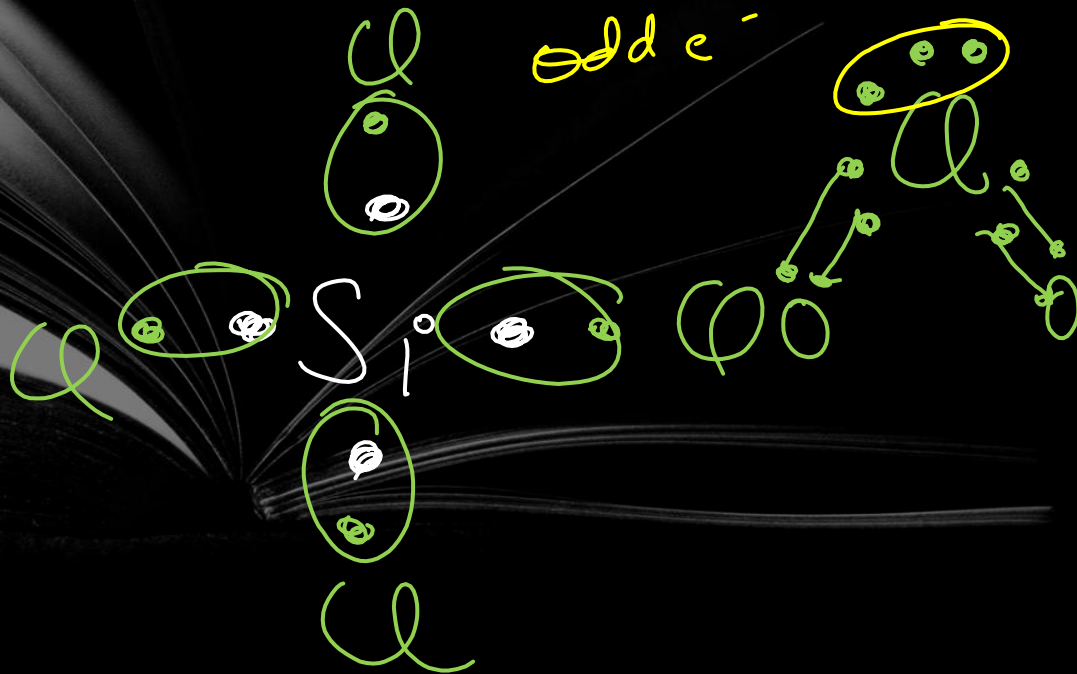
B



C



D



Which is not an exception to octet rule?

A



B



C



D



Which among the following chemical bond were described by Kossel and Lewis?

A

Metallic bond

B

Polar covalent bond

C

Coordinate bond

D

Ionic and Covalent bond

Which among the following chemical bond were described by Kossel and Lewis?

A

Metallic bond

B

Polar covalent bond

C

Coordinate bond

D

Ionic and Covalent bond

The octet rule is not obeyed

Homework

A



B



C

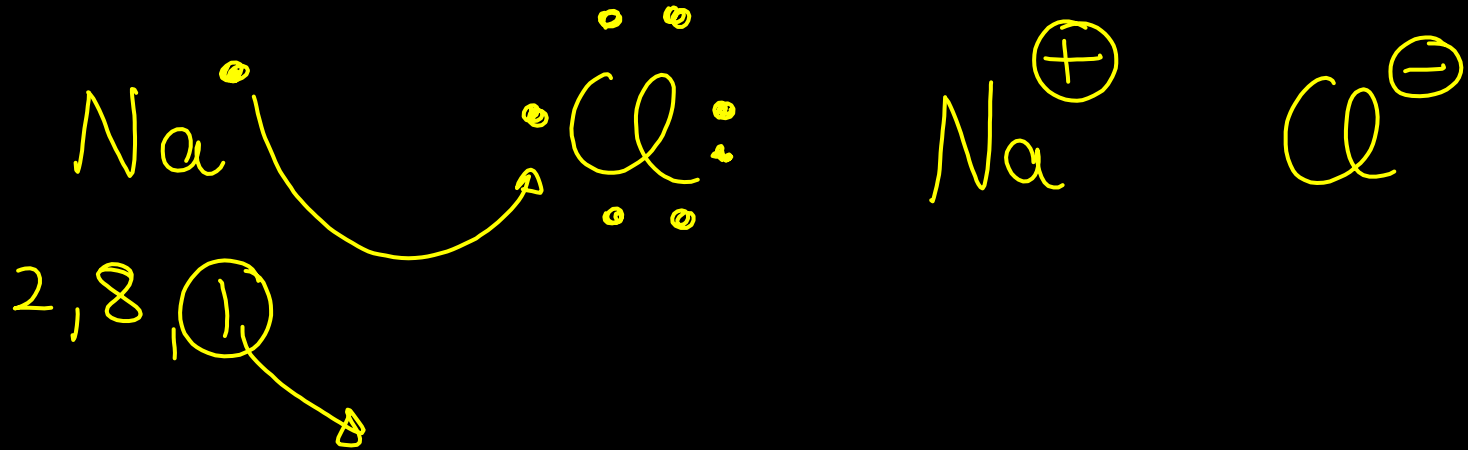


D



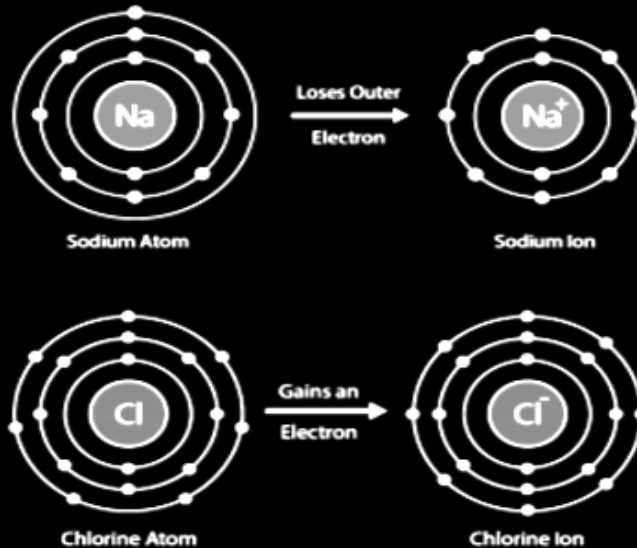
IONIC BOND

Electrovalent Bond



Electrovalent or Ionic Bonding

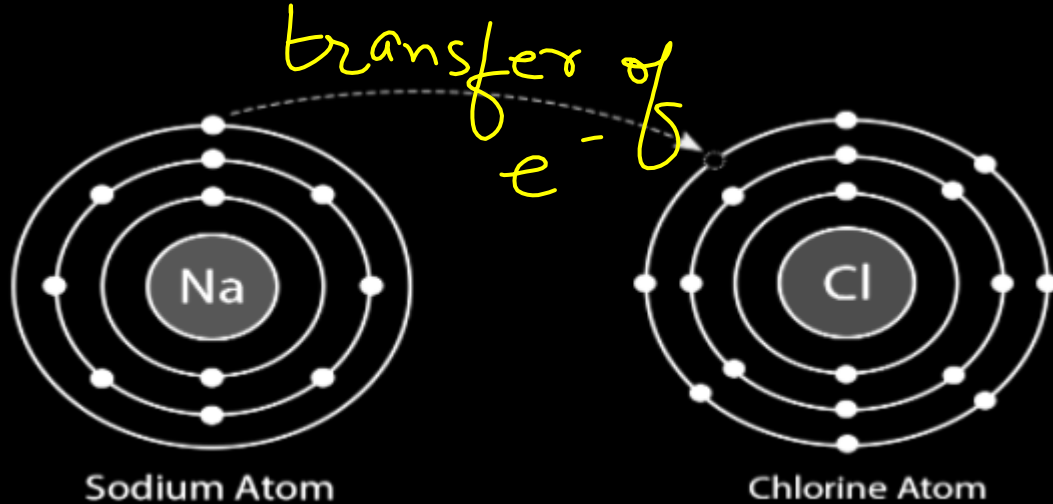
Ionic bonds are formed by loss and gain of e^- between two atoms.



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Electrovalent or Ionic Bonding

The electrostatic force of attraction developed between the cation and anion is known as ionic bond.

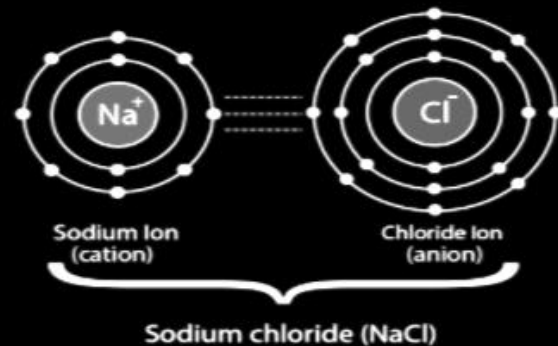
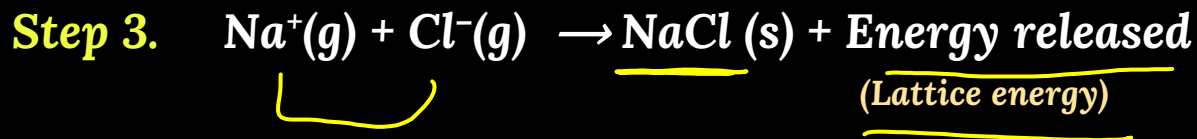
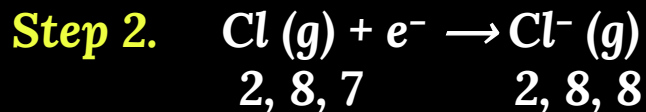
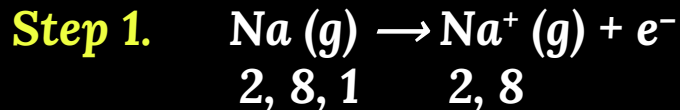


Metal

Non-metal

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Electroversion of Tumor Binding



3d arrangement
↓

Lattice energy

Conditions for Electrovalent or Ionic Bonding

```
graph TD; A[Conditions for Electrovalent or Ionic Bonding] --> B[Low ionisation energy of one atom]; A --> C[High electron affinity of other atom]; A --> D[High Lattice energy of compound formed];
```

✓
**Low ionisation
energy of one
atom**

✓
**High electron
affinity of
other atom**

**High Lattice
energy of
compound
formed**

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Lattice Energy

“The amount of energy released when free ions combine together to form one mole of a crystal is called lattice energy”

Higher the lattice energy greater will be the ease of formation of the ionic compound.



The crystal lattice of a sodium chloride molecule

Lattice Energy

$$\text{Lattice energy (L.E.)} \propto \frac{q_1 q_2}{r}$$

$r = r^+ + r^-$ (interionic distance)

r^+ = radius of cation

r^- = radius of anion

q_1 = charge on cation

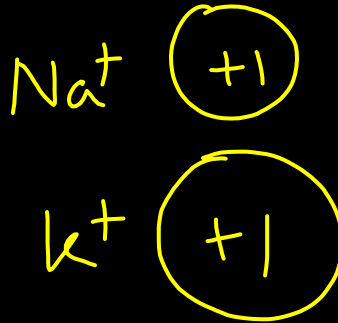
q_2 = charge on anion

Factors affecting lattice energy

Size of ions

$$\text{Lattice Energy} \propto \frac{1}{\text{Size of ions}}$$

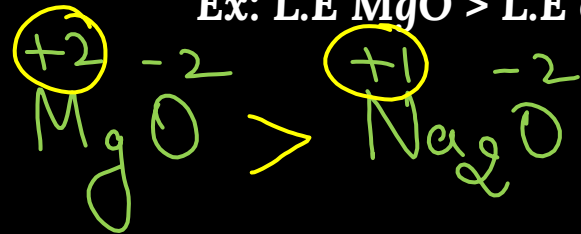
Example: $\text{NaCl} > \text{KCl}$



Charge on ions

$$\text{Lattice Energy} \propto \text{Charge on ions}$$

Ex: $\text{L.E MgO} > \text{L.E of Na}_2\text{O}$

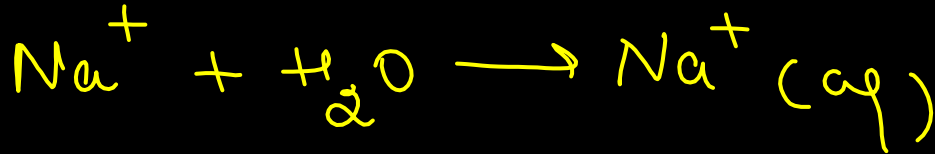


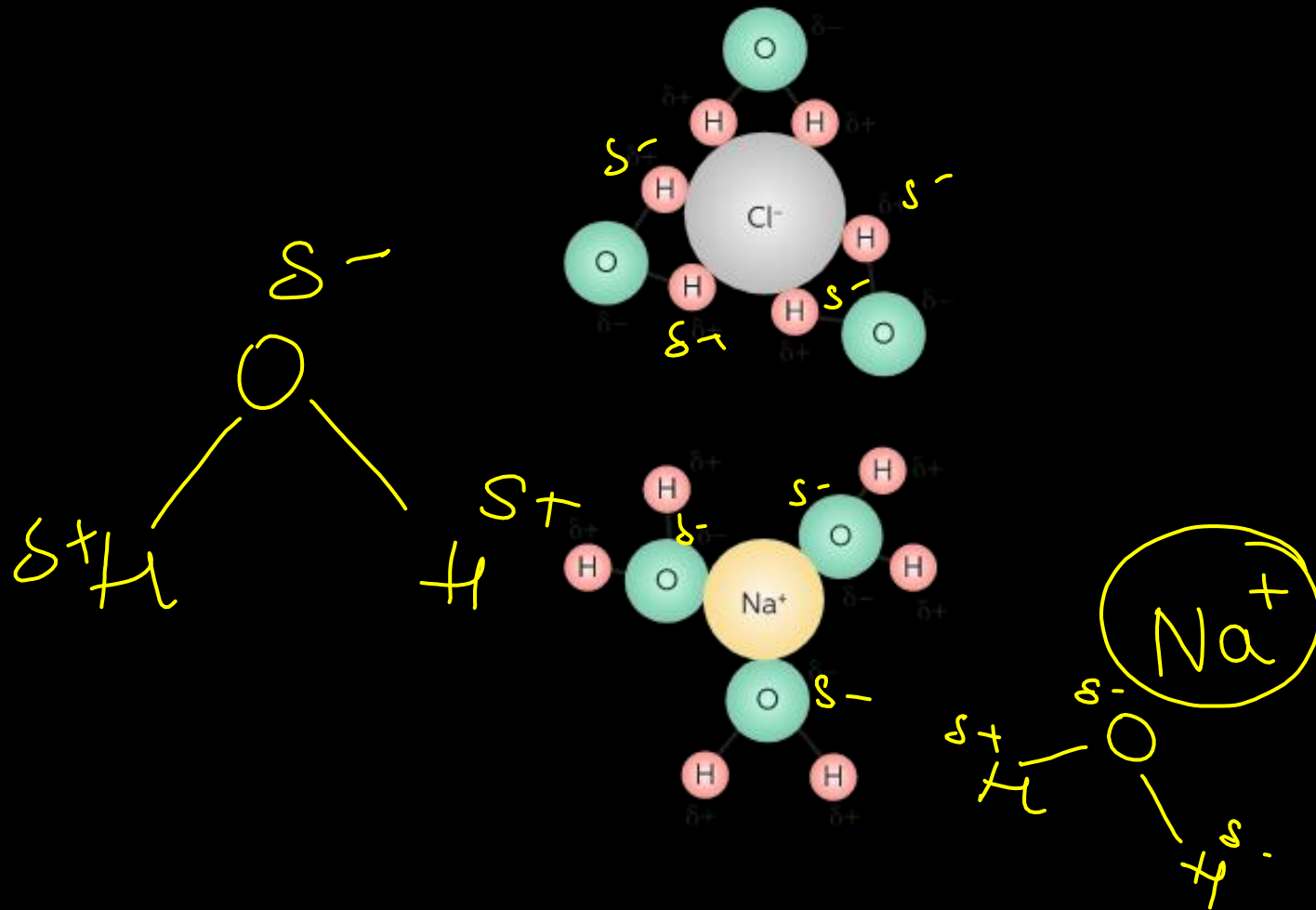
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Hydration Energy

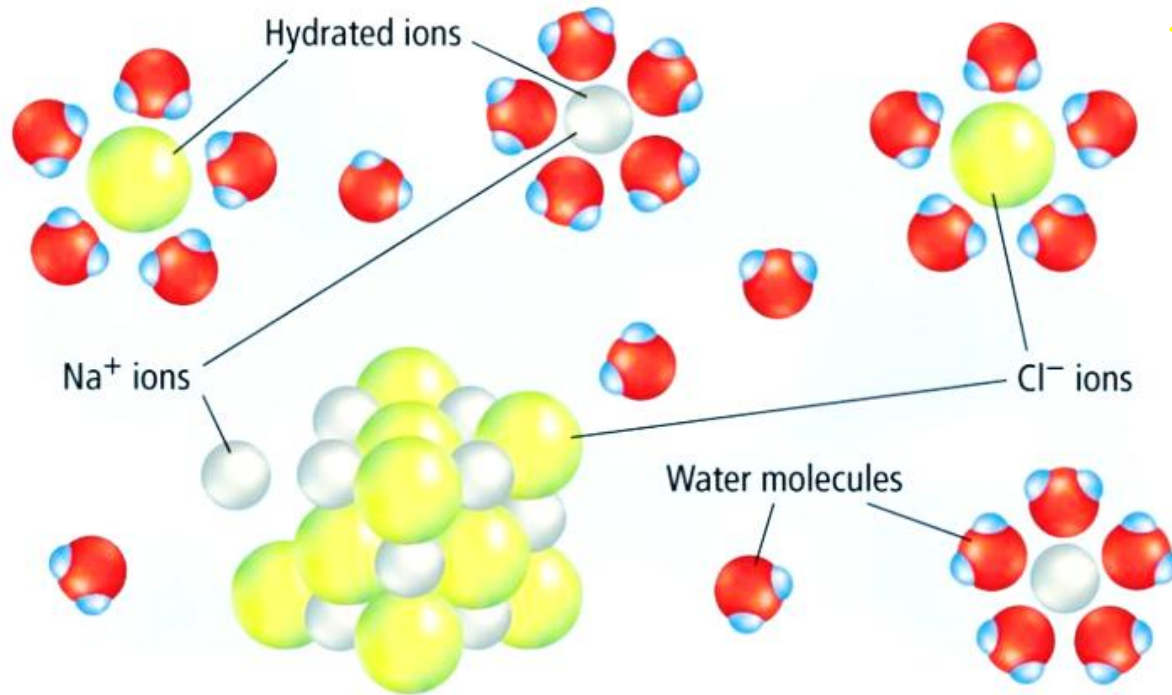
Whenever any compound is dissolved in a solvent or in water then, different ions of the compound will get separated and surrounded by solvent molecules. This process is known as solvation or hydration.

Energy released in this process is known as solvation energy or hydration energy.





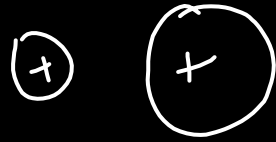
Solvation Process of NaCl



Na^+

Mg^{2+}

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Hydration $\propto \frac{1}{\text{Ionic mobility}}$

More is the charge density
 $\Downarrow$

More is the solvation / hydration energy
 Ionic mobility

$\text{Li}^+ > \text{Na}^+$

$\text{Li}(\text{aq}) < \text{Na}^+(\text{aq})$

Characteristics of ionic compounds

Physical state

Solid



High

Melting & boiling point

Electrical conductivity

molten/
solution

Solubility of
Ionic compound

Like
dissolves
like
polar
solvent

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Solubility

$$\text{Lattice energy} \propto \frac{1}{\text{Solubility}}$$

$$\text{Hydration energy} \propto \text{Solubility}$$

NO Bond is 100% Ionic

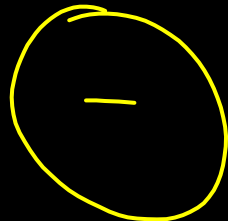
Na

Na^+



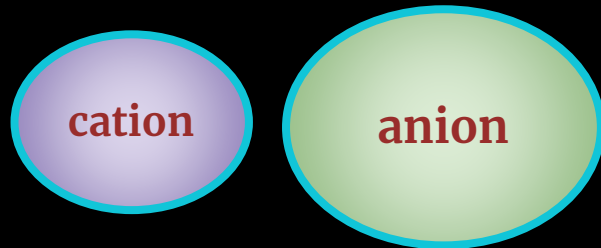
Cl

Cl^-

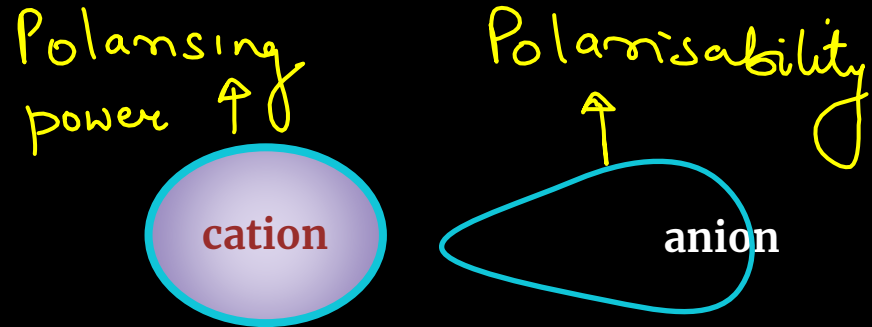


Fajan's Rule

- 👉 Each bond has mix character of covalent and Ionic.
- 👉 When two oppositely charged ions of unequal size approach each other closely (during formation of an electrovalent bond), the ion smaller in size attracts outermost electrons of the other ion and repel its nuclear charge.



(a) No Polarisation



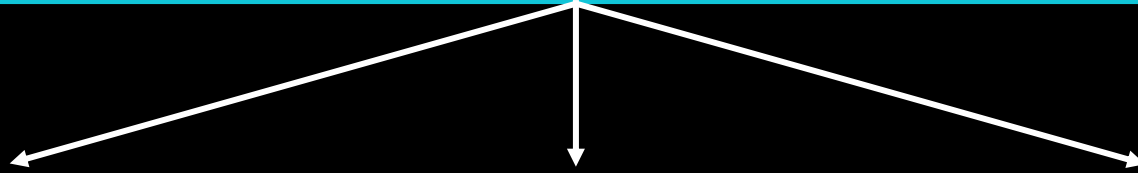
(b) Polarised anion

Fajan's Rule

- 👉 The ability of a cation to polarise the nearby anion is called its polarising power.
- 👉 Tendency of an anion to get distorted or deformed or polarised by the cation is called its polarisability.
- 👉 Due to polarisation, sharing of electrons occur between two ions to some extent and the bond shows some covalent character.

Polarisation & covalent character

Factors affecting polarising power (cations)



Charge on ions

more

Size of the ions

→ Cation
↓

Smaller size



Electronic Configuration of cation

Pseudo inert gas conf

Fajan's Rule

Covalent $\propto$ Polarisation

$\propto \frac{1}{r}$

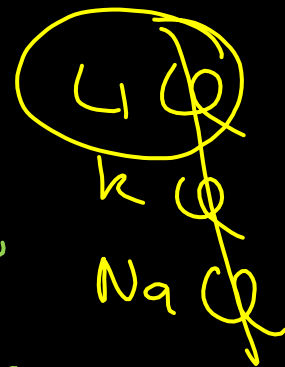
Ionic
✓ character



Small cation

Anion, big
charge $\times$ on ion

pseudo inert gas



Factors affecting polarising power (cations)

Charge on ions

Polarising power $\propto Z_{\text{eff}} \propto \text{charge } (Z^+)$

Polarising power $\uparrow$ ionic character $\downarrow$

Example: $\text{Li}^+ < \text{Be}^{+2} < \text{Al}^{+3}$

Factors affecting polarising power (cations)

Size of the ions

✓
$$\text{Polarising power} \propto \frac{1}{\text{size}}$$

Polarising power $\uparrow$ ionic character $\downarrow$

Example: (i) $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$

(ii) $\text{Be}^{+2} > \text{Mg}^{+2} > \text{Ca}^{+2}$

Factors affecting polarising power (cations)

Electronic Configuration of cation

Polarising power $\propto$ Pseudo inert gas configuration ($ns^2np^6nd^{10}$)

Example: $\begin{matrix} \text{Na}^+ & \text{K}^+ & \text{Mg}^{+2} & \text{Ca}^{+2} \end{matrix} \left\{ \begin{matrix} \text{2,8} & \text{2,8,8} & \text{2,8} & \text{2,8,8} \end{matrix} \right\}$ Noble gas

Cu^+
2, 8, 18 ✓

(Note: More polarising power if the cations have pseudo noble gas configuration)

Factors affecting polarisation (Anions)

Charge of ion ✓

Polarisation $\propto$ charge (Z^-)

Example:

- (i) $F^- < O^{-2} < N^{-3}$
- (ii) $Cl^- < S^{-2} < P^{-3}$
- (iii) $NO_2^- < NO_3^-$
- (iv) $SO_3^{-2} < SO_4^{-2}$

Size of the ion

Polarisation $\propto$ size (r^-)

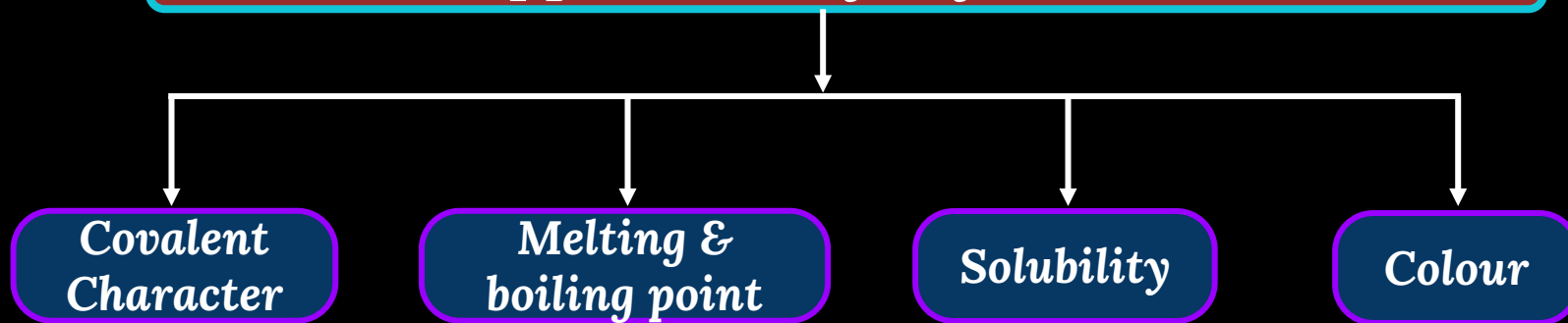
Example:

- (i) $F^- < Cl^- < Br^- < I^-$
- (ii) $O^{-2} < S^{-2} < Se^{-2}$
- (iii) $N^{-3} < P^{-3}$

Note: If the size and charge contradict each other than charge dominates.

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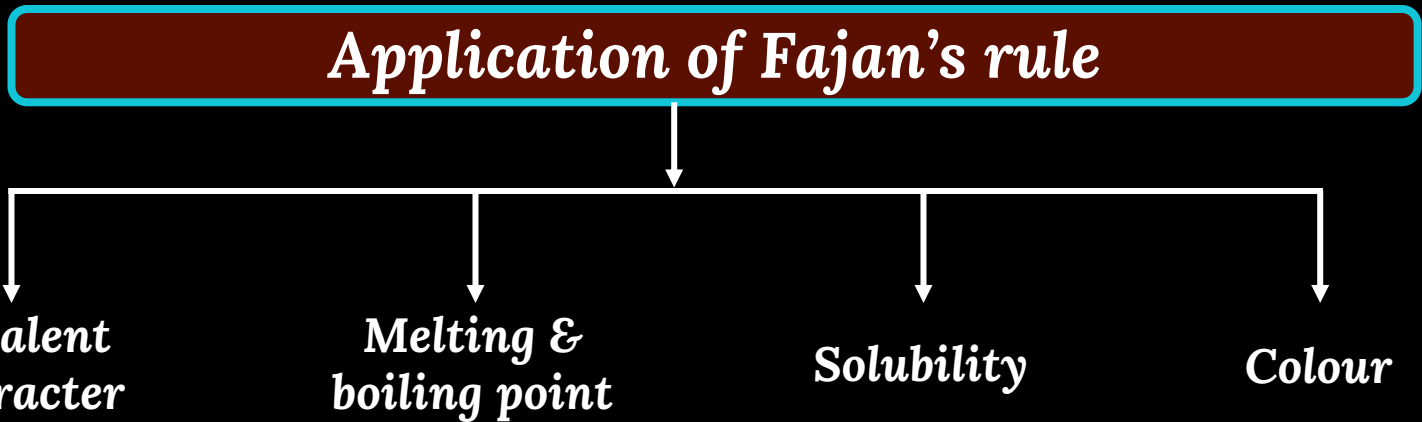
Application of Fajan's rule



👉 $\text{Polarising power} \propto \text{Polarisation} \propto \text{Covalent character} \propto \frac{1}{\text{M.P. and B.P.}}$

👉 $\text{Covalent Character} \propto \frac{1}{\text{Solubility}} \propto \text{colour imparting tendency}$

Application of Fajan's rule



```
graph TD; A[Application of Fajan's rule] --> B[Covalent Character]; A --> C[Melting & boiling point]; A --> D[Solubility]; A --> E[Colour]
```

**Covalent
Character**

**Melting &
boiling point**

Solubility

Colour

- 👉 Compound in which ionic Character less than 20% are found to be coloured whether they contain unpaired electron or not.
- 👉 Larger is the size of anion more easier will be the deformation of electron cloud.
- 👉 Easier excitation of electron so greater will be the intensity of colour.

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An ionic compound $A^+ B^-$ is most likely to be formed when :

A

the ionization energy of A high and electron affinity of B is low

B

the ionization energy of A is low and electron affinity of B is high

C

both, the ionization energy of A and electron affinity of B are high

D

both, the ionization energy of A and electron affinity of B are low

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An ionic compound $A^+ B^-$ is most likely to be formed when :

A

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B

the ionization energy of A is low and electron affinity of B is high

C

both, the ionization energy of A and electron affinity of B are high

D

both, the ionization energy of A and electron affinity of B are low

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Telegram Channel



 **Don't Miss Out!**



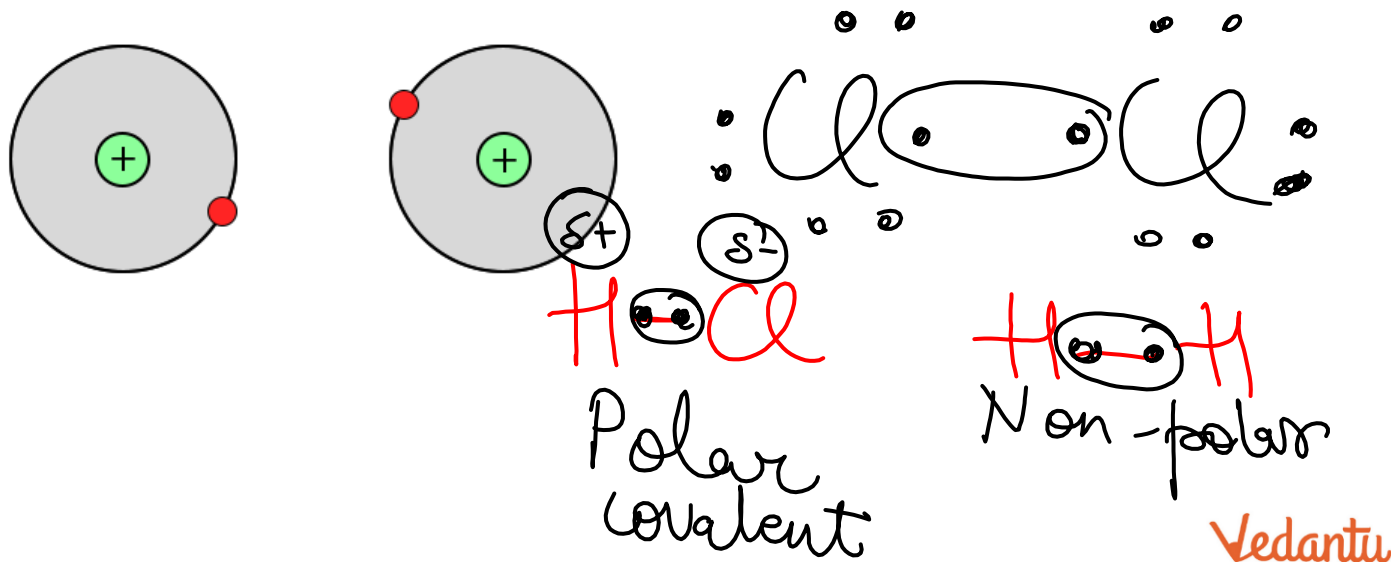
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Hit **LIKE**, **SHARE**, and **SUBSCRIBE**



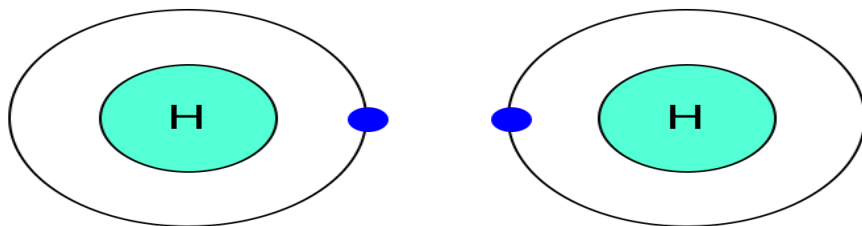
COVALENT BOND

A covalent bond may be defined as the bond formed by mutual sharing of electrons between the combining atoms of comparable Electronegativity.



Conditions for formation of covalent bonds

- An atom will not transfer the electron or electrons to the other atom if the electronegativity difference between the two atoms is zero or very small (less than 1.6).
- When both the atoms are short in electrons in the valence shell in comparison to stable noble configuration, then such atoms complete their valence shells by sharing electrons.



Characteristics of covalent compounds



Physical state

gas/liquid

I_2, P_4 , diamond & except
solid 3d-network

Melting and boiling points

Low

Electrical conductivity

Bad

except graphite

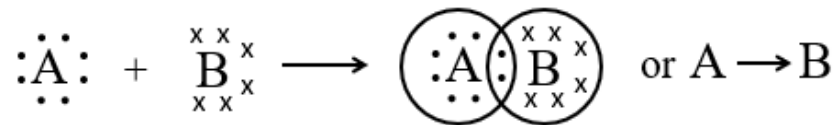
Solubility

Polar & Polar
N.Polar & N.P

HCl in water

Coordinate Bond

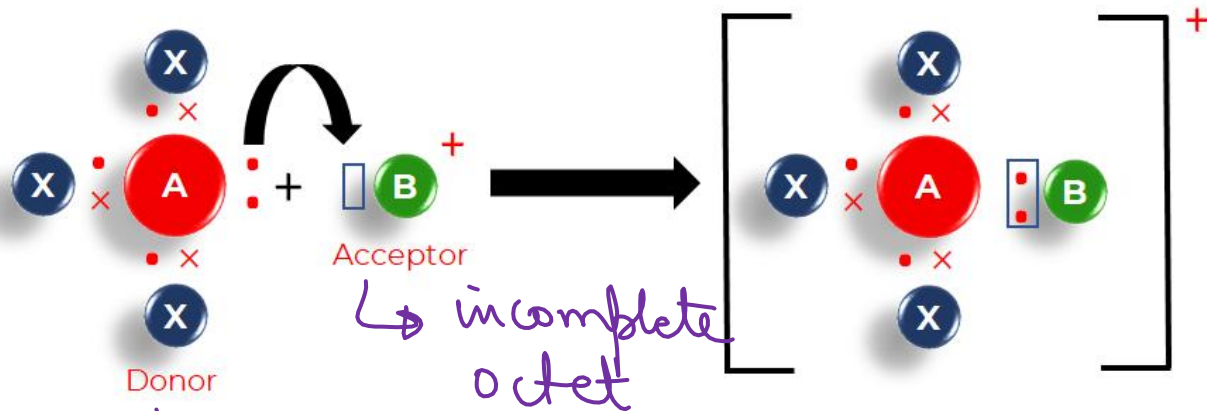
- “A covalent bond in which both electrons of the shared pair are contributed by one of the two atoms”. Such a bond is also called as dative bond.
- A coordinate or a dative bond is established between two such atoms, one of which has a complete octet and possesses a pair of valence electrons while the other is short of a pair of electrons.



- The atom which contributes electron pair is called the donor and also known as Lewis base while the atom which accepts it is called acceptor and also known as Lewis acid.

Coordinate Bond

- Formed between two atom in which contribution of an e^- pair is made by one of them while the sharing is done by both.



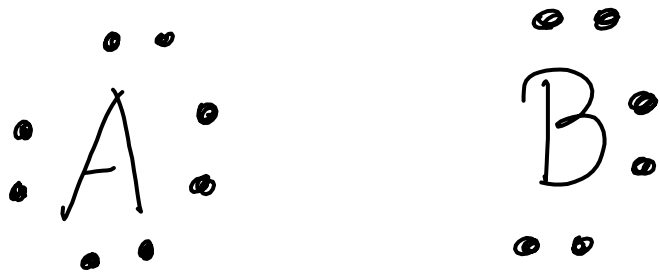
Important

Adduct



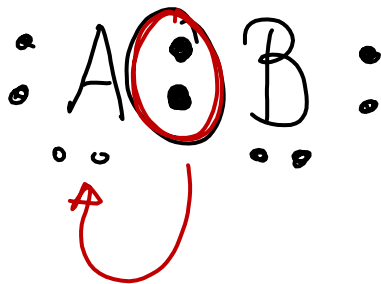
Donor

Acceptor



Donor

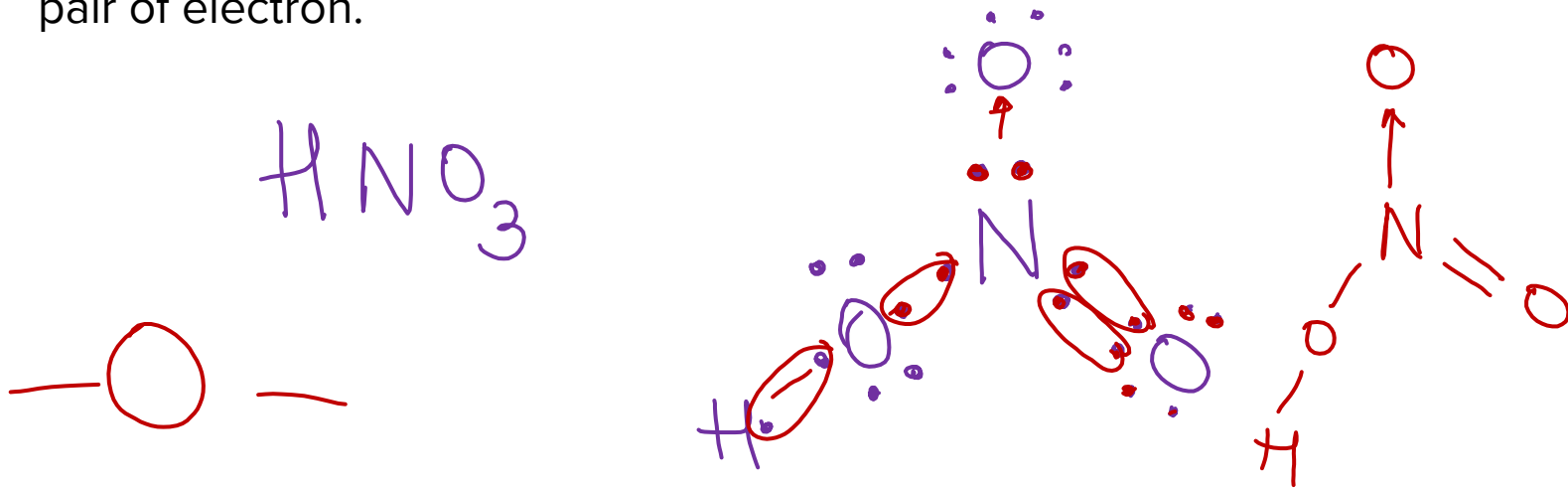
Acceptor



Co-ordinate /
Dative Bond

Coordinate Bond

- ❖ Necessary conditions for the formation of Coordinate bond are -
 - Octet of one atom should be complete and should have at least one lone pair of electron.



Covalency

It is defined as number of bonds formed by an atom in the compound as number of electron shared by the atom in the compound.

Example:



(i) BF_3 or BF_4^- $\longrightarrow$ Boron Covalency 3 or 4

(ii) H_2O or H_3O^+ $\longrightarrow$ Oxygen Covalency 2 or 3

(iii) NH_3 or NH_4^+ $\longrightarrow$ Nitrogen Covalency 3 or 4

VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY :



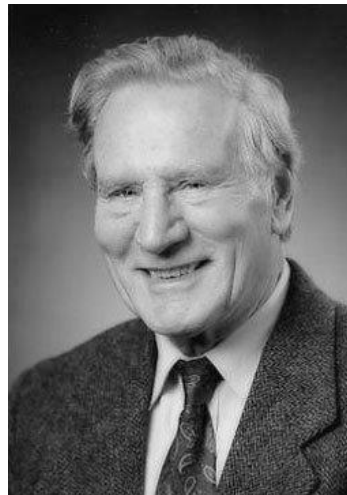
Ronald Sydney Nyholm

V V Imp

Shapes



Covalent molecule

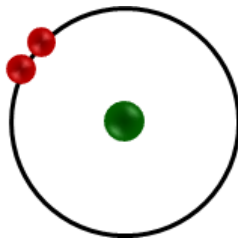


Ronald Gillespie

To predict and explain molecular shapes and bond angles properly

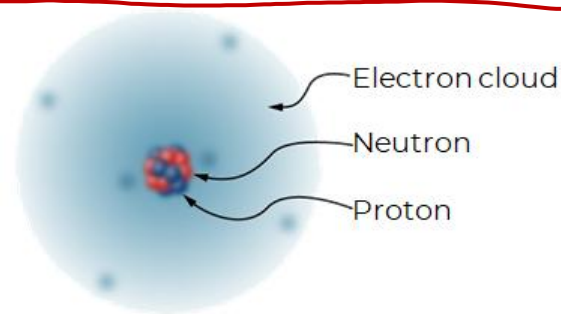
Postulates of VSEPR

- The shape of a molecule depends upon the number of valence shell e^- pairs (bonded or unbonded) around the central atom.



Postulates of VSEPR

- The shape of a molecule depends upon the number of valence shell e^- pairs (bonded or unbonded) around the central atom.
- Pairs of e^- in the valence shell repel one another since their e^- clouds are negatively charged.



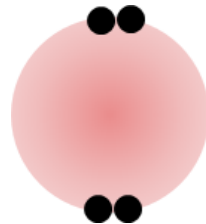
Postulates of VSEPR

- The shape of a molecule depends upon the number of valence shell e^- pairs (bonded or unbonded) around the central atom.
- Pairs of e^- in the valence shell repel one another since their e^- clouds are negatively charged.
- These pairs of e^- tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.



Postulates of VSEPR

- The valence shell is taken as a sphere with the e^- pairs localising on the spherical surface at maximum distance from one another.



- Multiple Bond

Treated as → Single electron pair

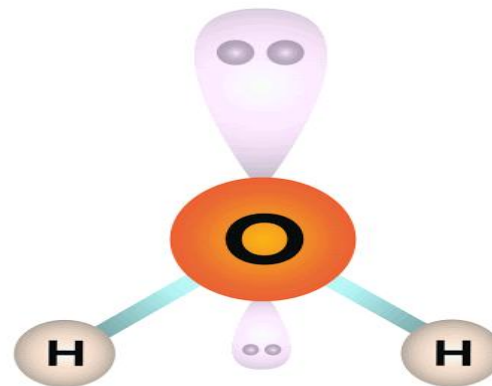
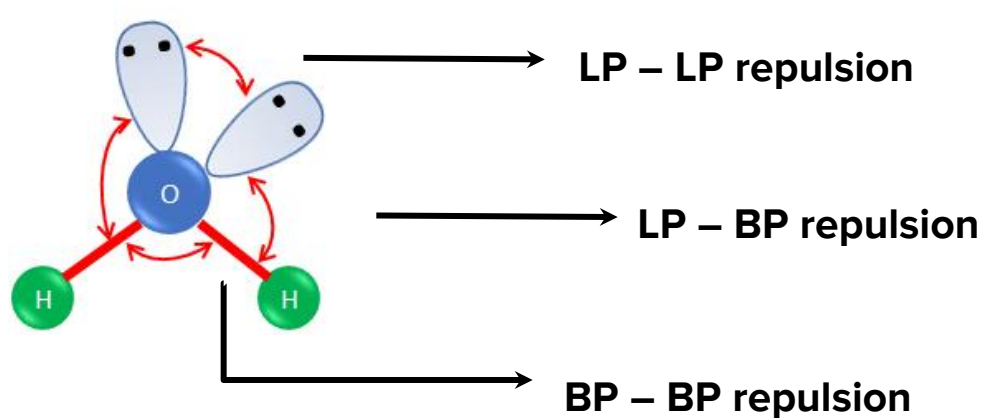
Two or three
electron pairs

Treated as → Single super pair

Order of repulsive Interaction

The repulsive interaction of electron pairs decreases in the order :

lone pair (lp) - lone pair (lp) > lone pair (lp) - bond pair (bp) > bond pair (bp) - bond pair (bp)

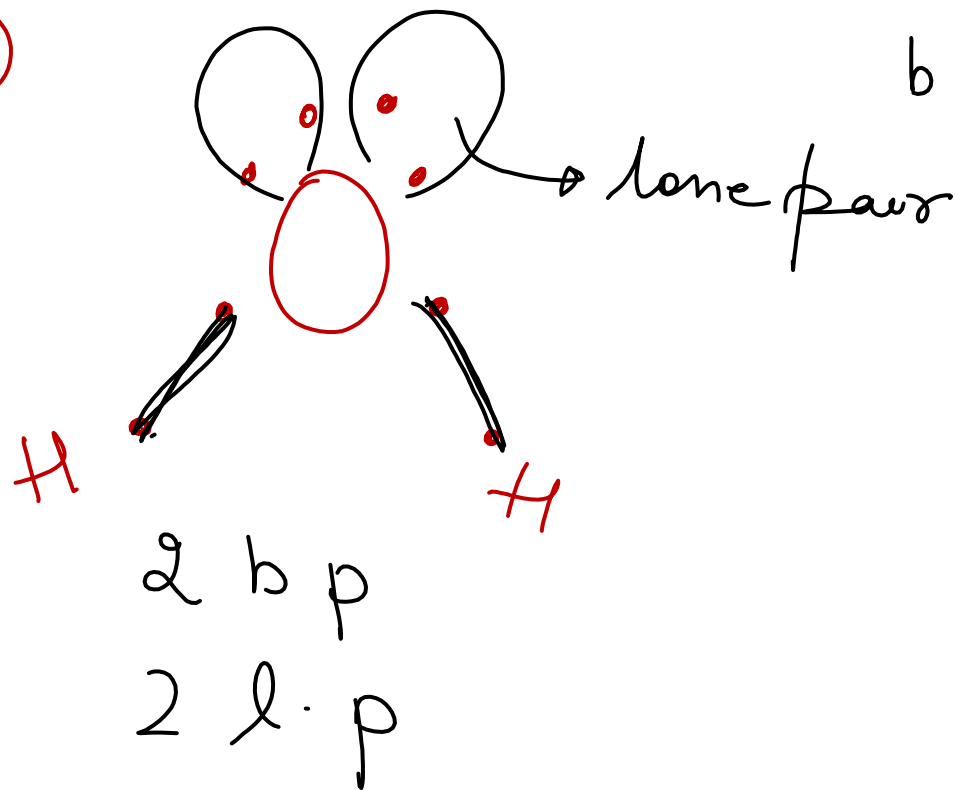
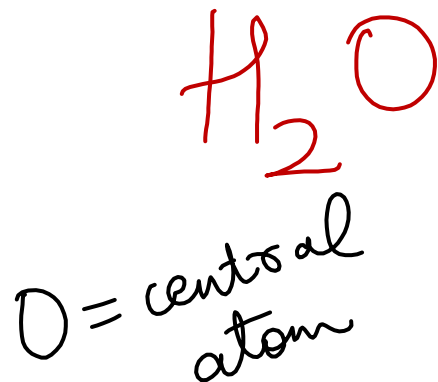


Prediction of geometrical shapes

For the prediction of geometrical shapes of molecules with the help of VSEPR theory it is convenient to divide molecules into two categories as

- molecules in which the central atom has **no lone pair**

- molecules in which the central atom / ion has **one or more lone pairs**



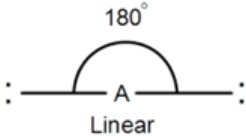
lone pair

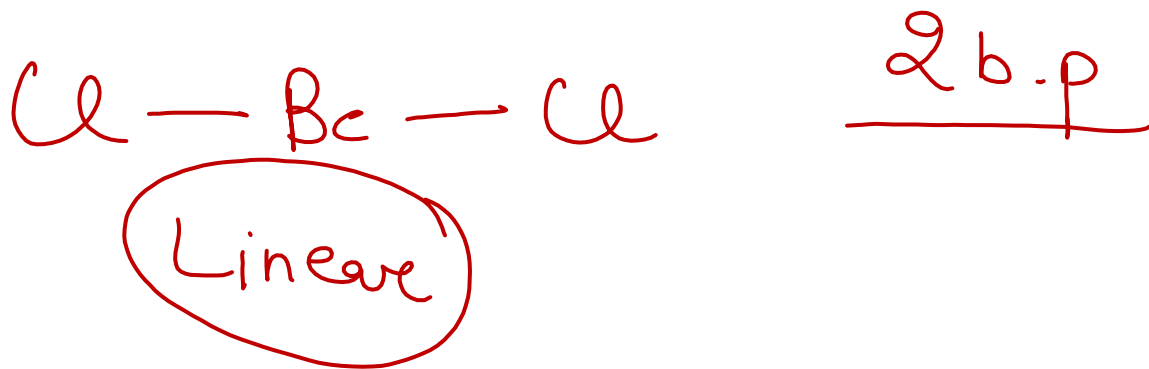
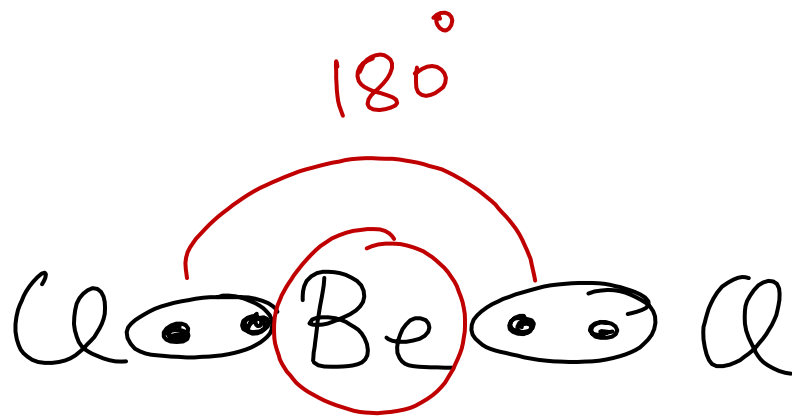
l p - l p

b p - b p

l p - b p

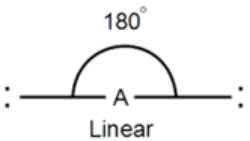
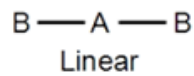
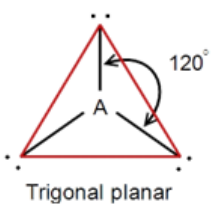
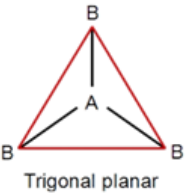
Geometry in which the central atom has no lp of e⁻

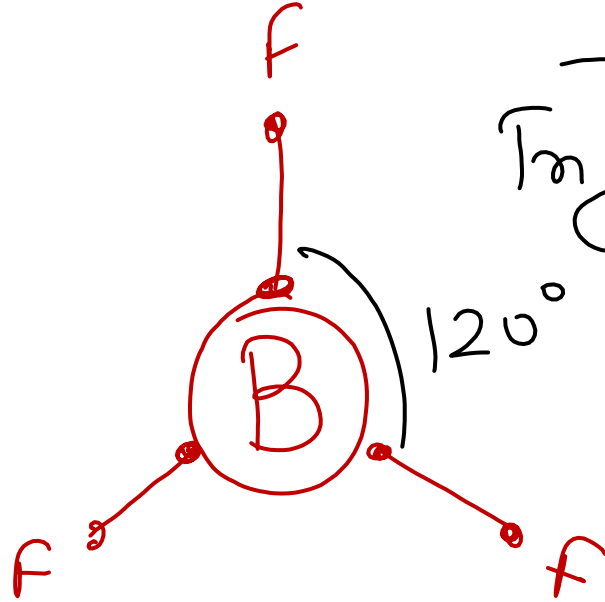
Number of electron pairs	Arrangement of electron pairs	Molecular Geometry	Examples
2		$B - A - B$ Linear	$BeCl_2$, $HgCl_2$





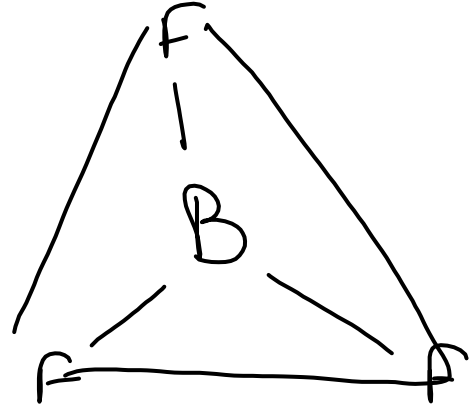
Geometry in which the central atom has no lp of e⁻

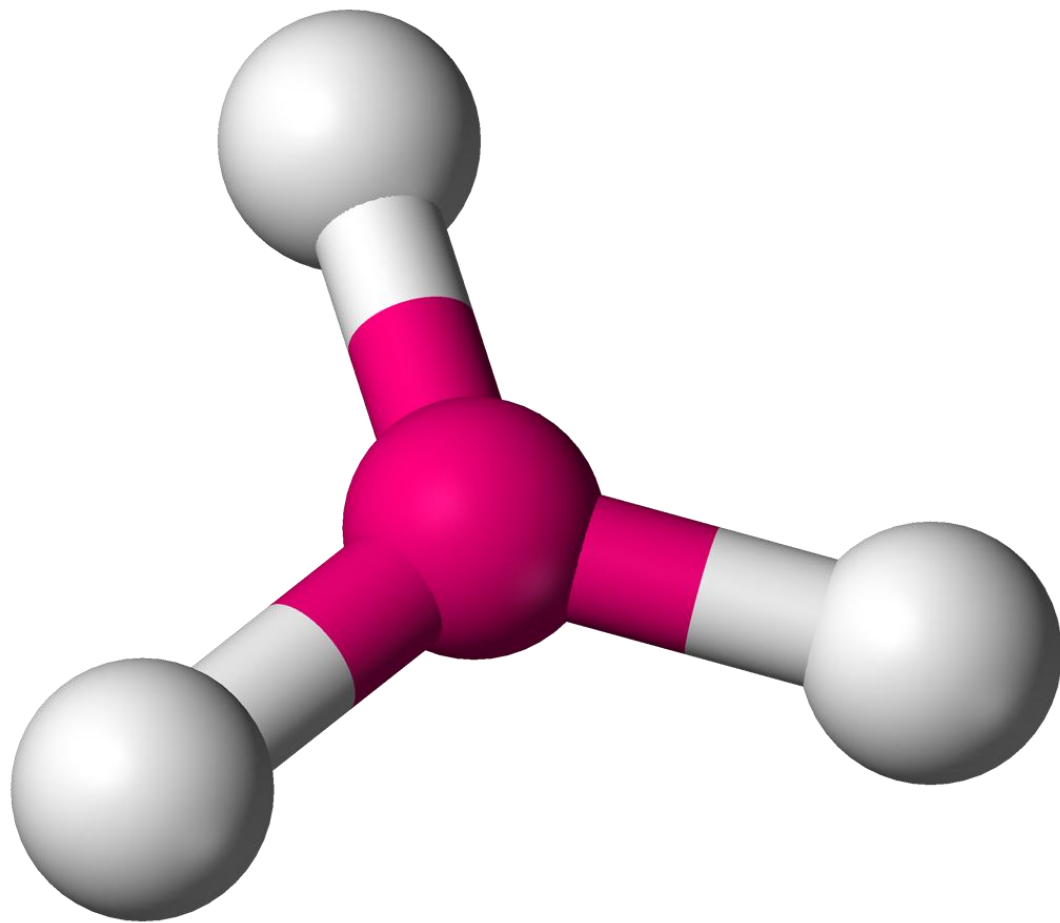
Number of electron pairs	Arrangement of electron pairs	Molecular Geometry	Examples
2			BeCl ₂ , HgCl ₂
3			BF ₃



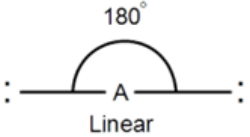
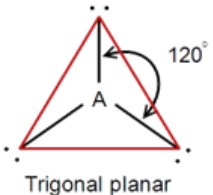
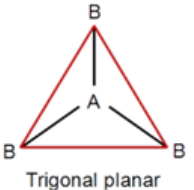
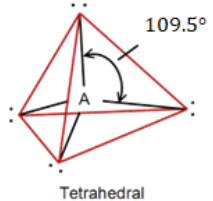
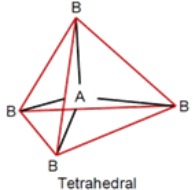
3 b p

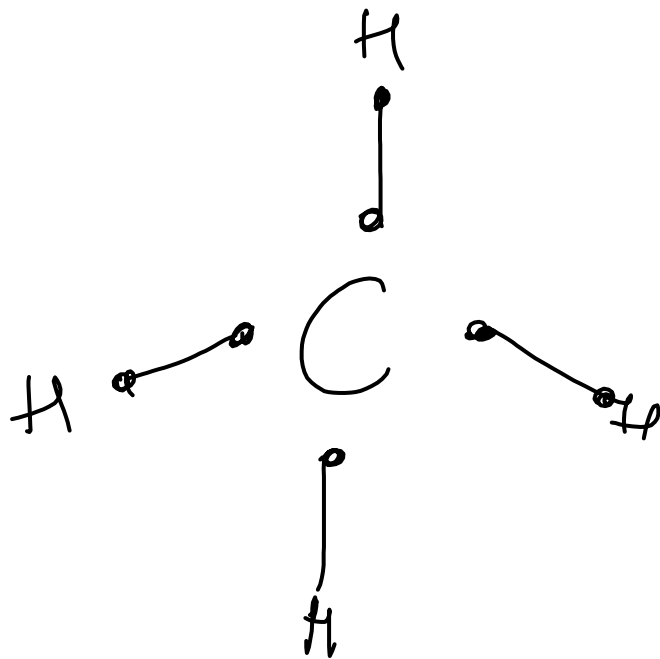
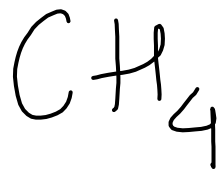
Trigonal Planar



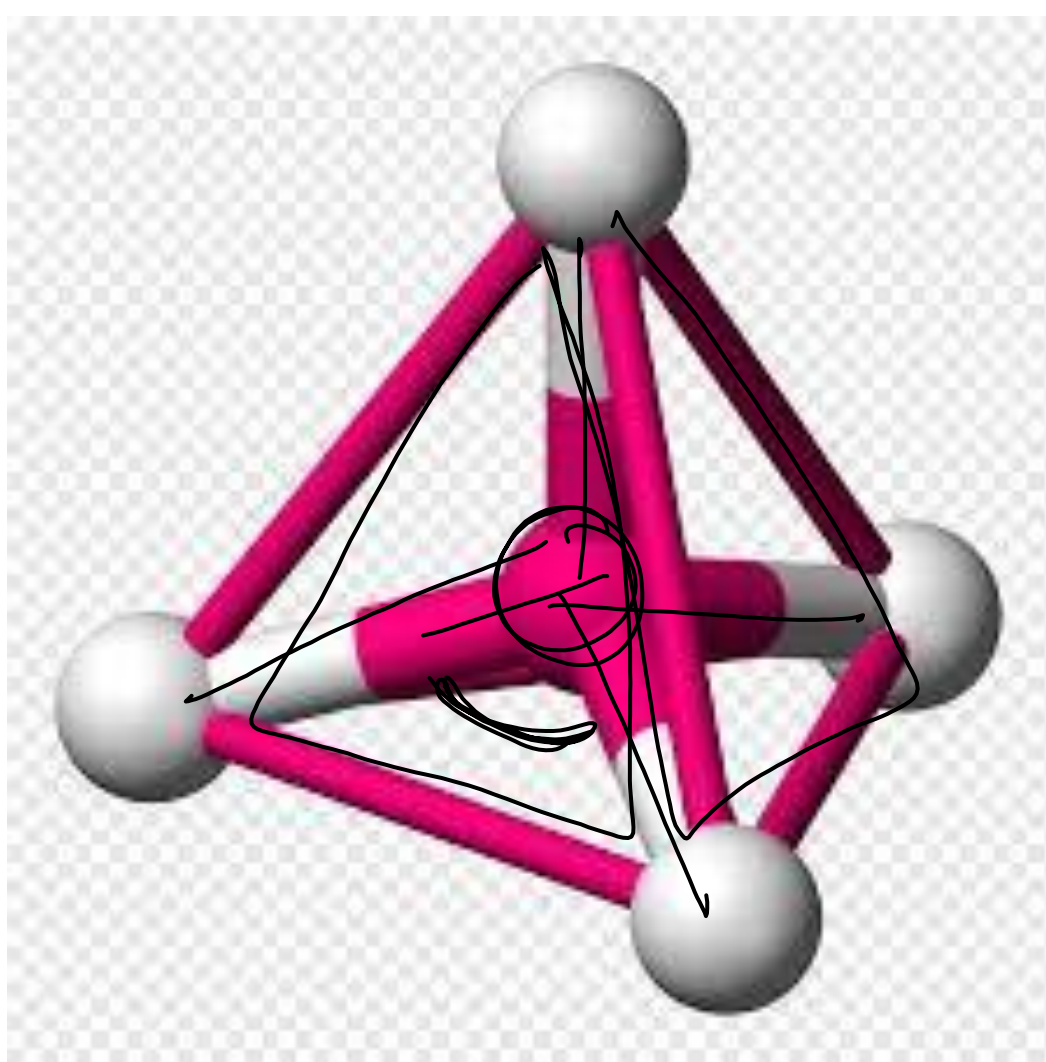


Geometry in which the central atom has no lp of e⁻

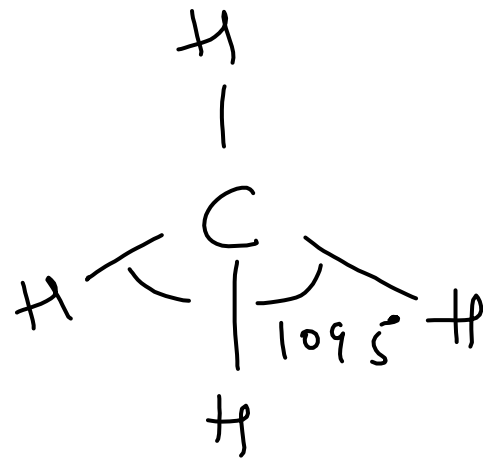
Number of electron pairs	Arrangement of electron pairs	Molecular Geometry	Examples
2		$B - A - B$ Linear	$BeCl_2$, $HgCl_2$
3			BF_3
4			CH_4 , NH_4^+



4 b p

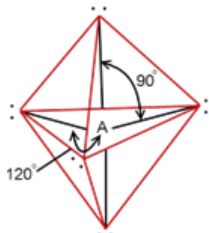
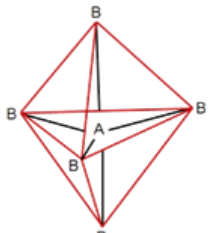


Tetrahedral



109.5°

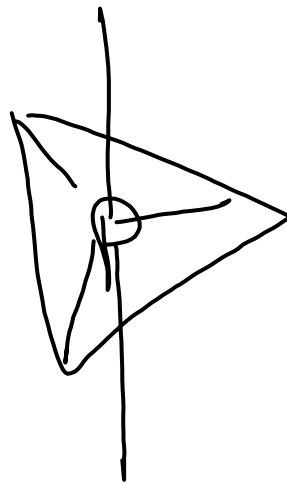
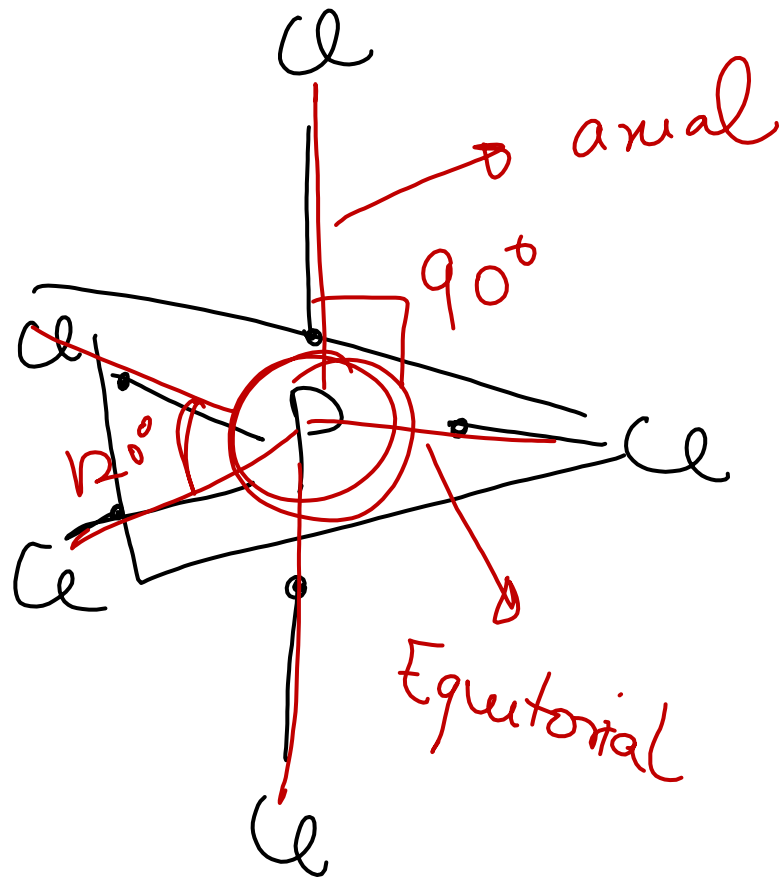
Geometry in which the central atom has no lp of e⁻

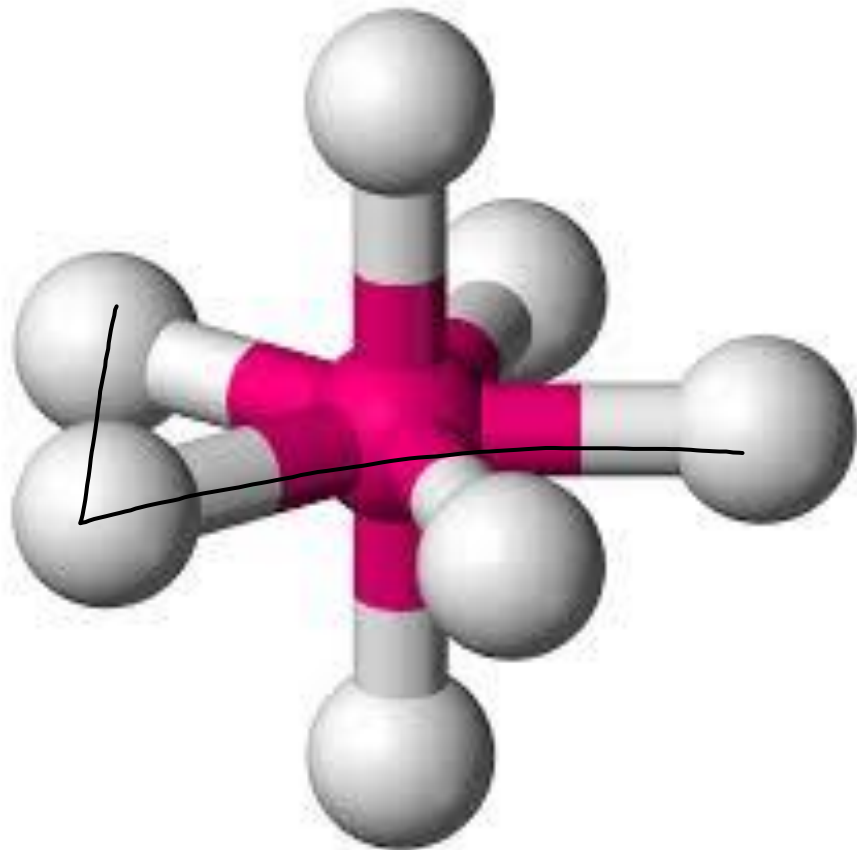
Number of electron pairs	Arrangement of electron pairs	Molecular Geometry	Examples
5	 Trigonal bipyramidal	 Trigonal bipyramidal	PCl₅

PCl_5

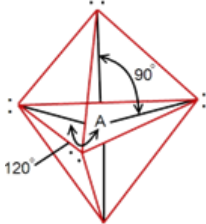
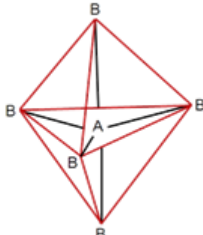
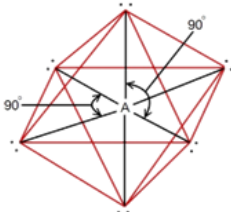
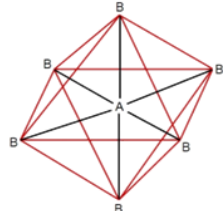
TBP

5 b p



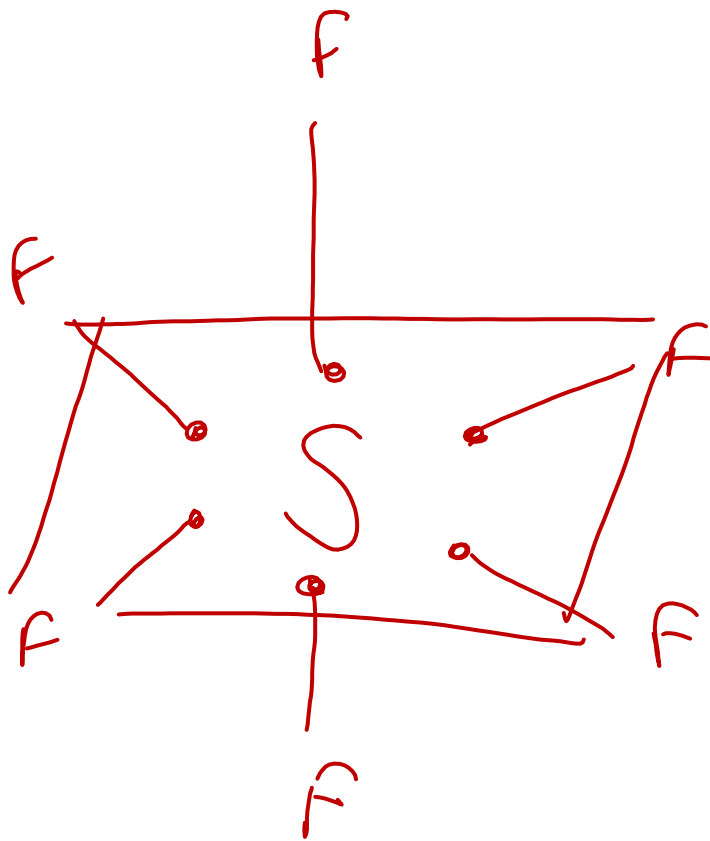


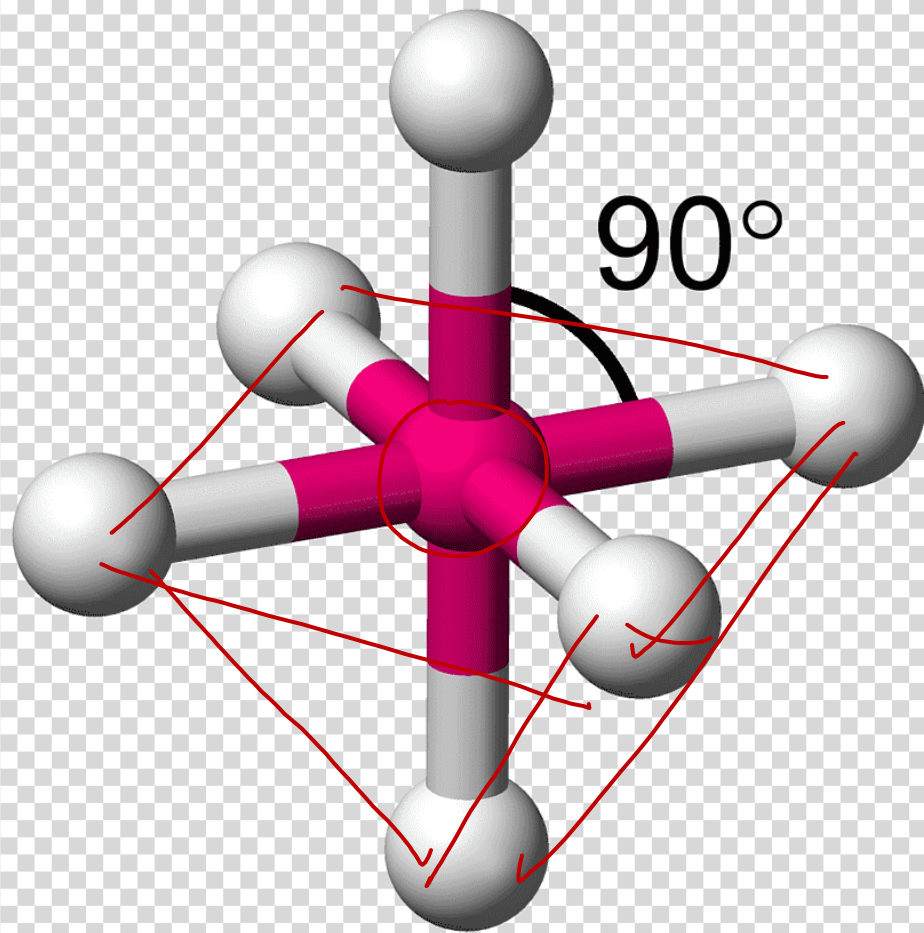
Geometry in which the central atom has no lp of e⁻

Number of electron pairs	Arrangement of electron pairs	Molecular Geometry	Examples
5	 Trigonal bipyramidal	 Trigonal bipyramidal	PCl₅
6	 Octahedral	 Octahedral	SF₆

6bp,

Sf₆



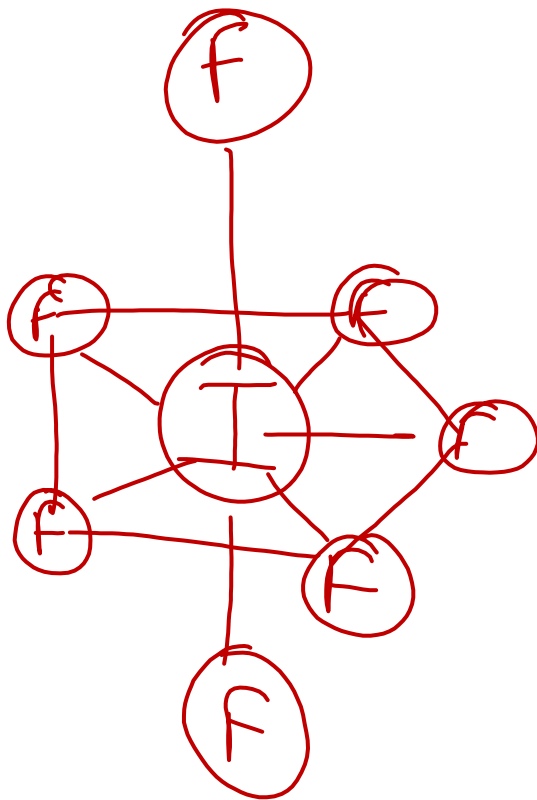


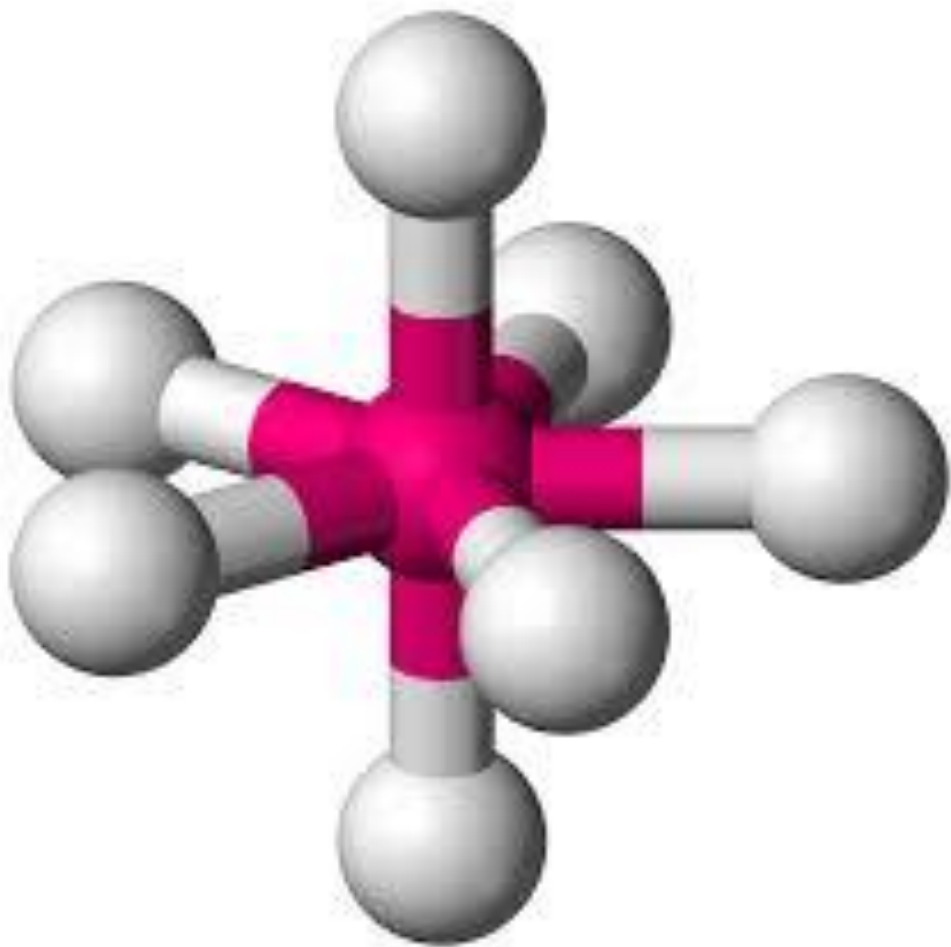
octahedral

7bp.

PBP

If₇







Summary

Geometry in which the central atom has no lp of e^- :

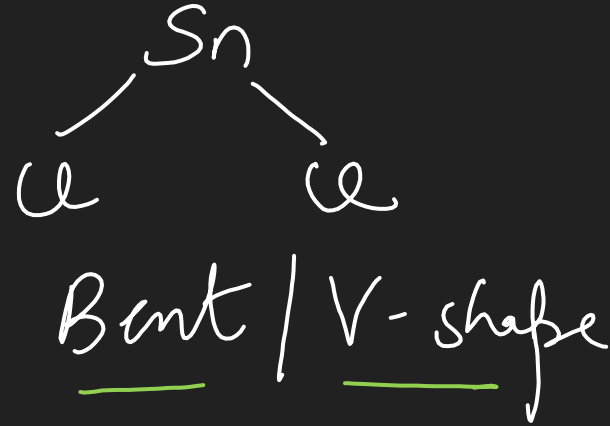
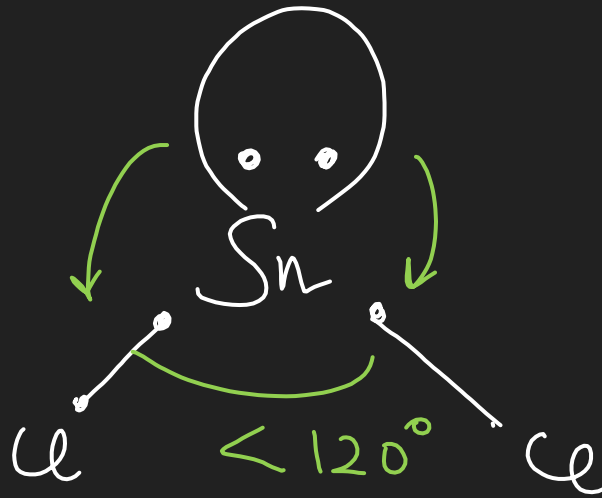
No. of e^- pairs	Molecular Geometry
2	Linear
3	Trigonal planar
4	Tetrahedral
5	Trigonal bipyramidal
6	Octahedral

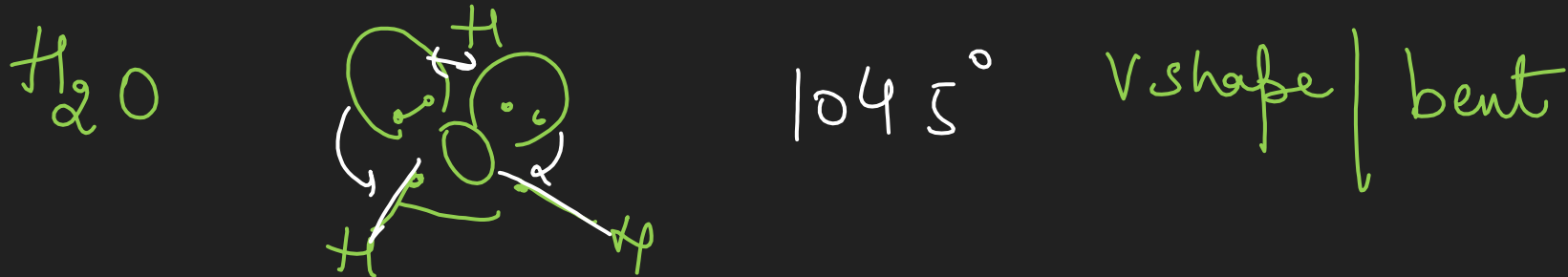
Handwritten notes: A red circle around the number 3 in the first column, with the letters "BP" written inside. A red bracket is drawn to the left of the numbers 3, 4, 5, and 6, spanning from the row for 3 down to the row for 6.

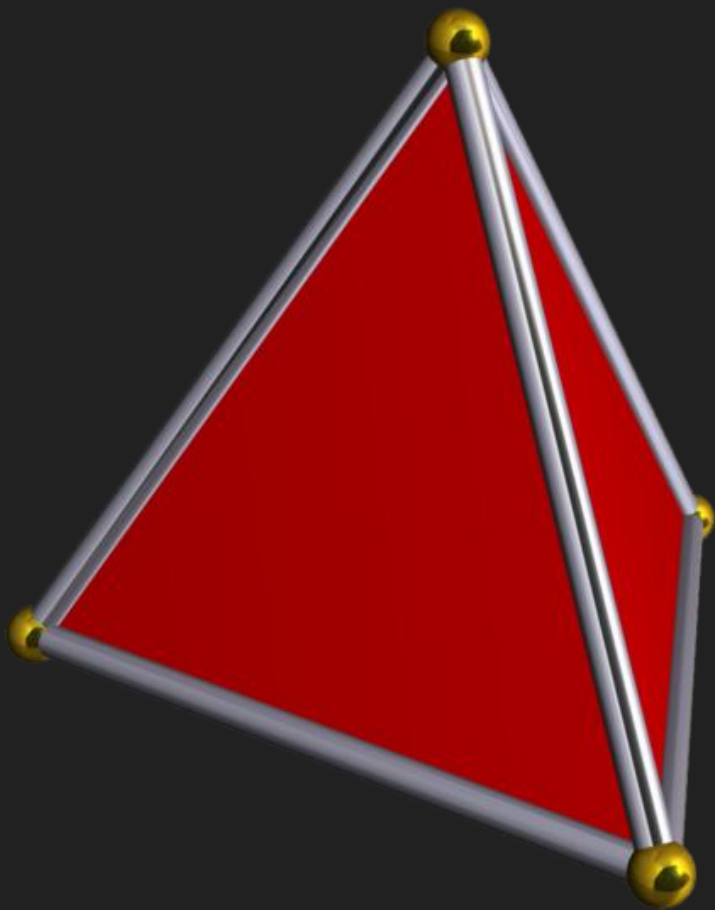
VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY



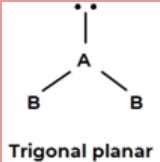
2 b p }
1 l p } (3)

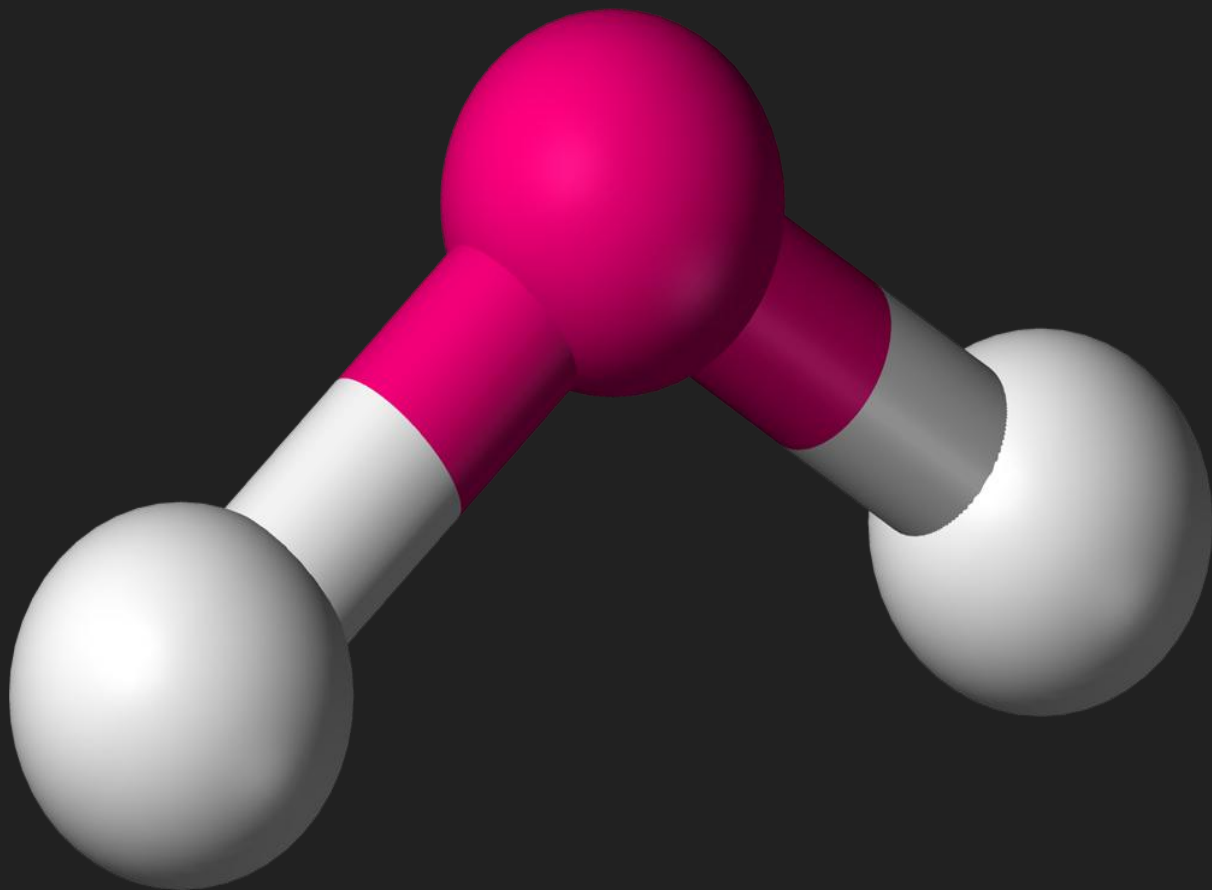




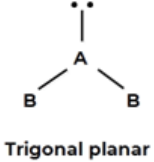
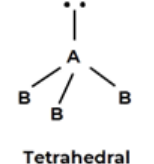


Geometry in which central atom has one or more lp of e⁻

Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB₂E	2	1	 Trigonal planar	Bent, SO₂, O₃



Geometry in which central atom has one or more lp of e⁻

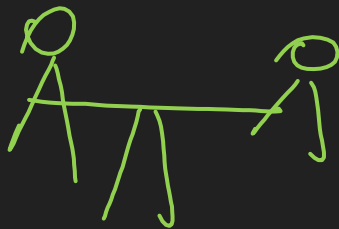
Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_2E	2	1	 Trigonal planar	Bent, SO_2 , O_3
AB_3E	3	1	 Tetrahedral	Trigonal Pyramidal, NH_3

PC_5

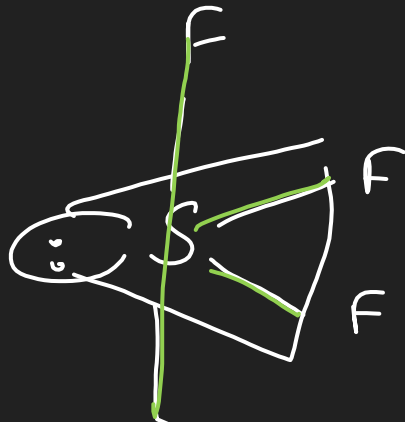
TBP

5bp, 0lp

SF_4

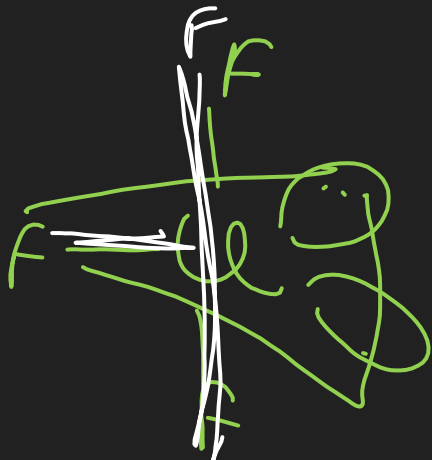


UF_3



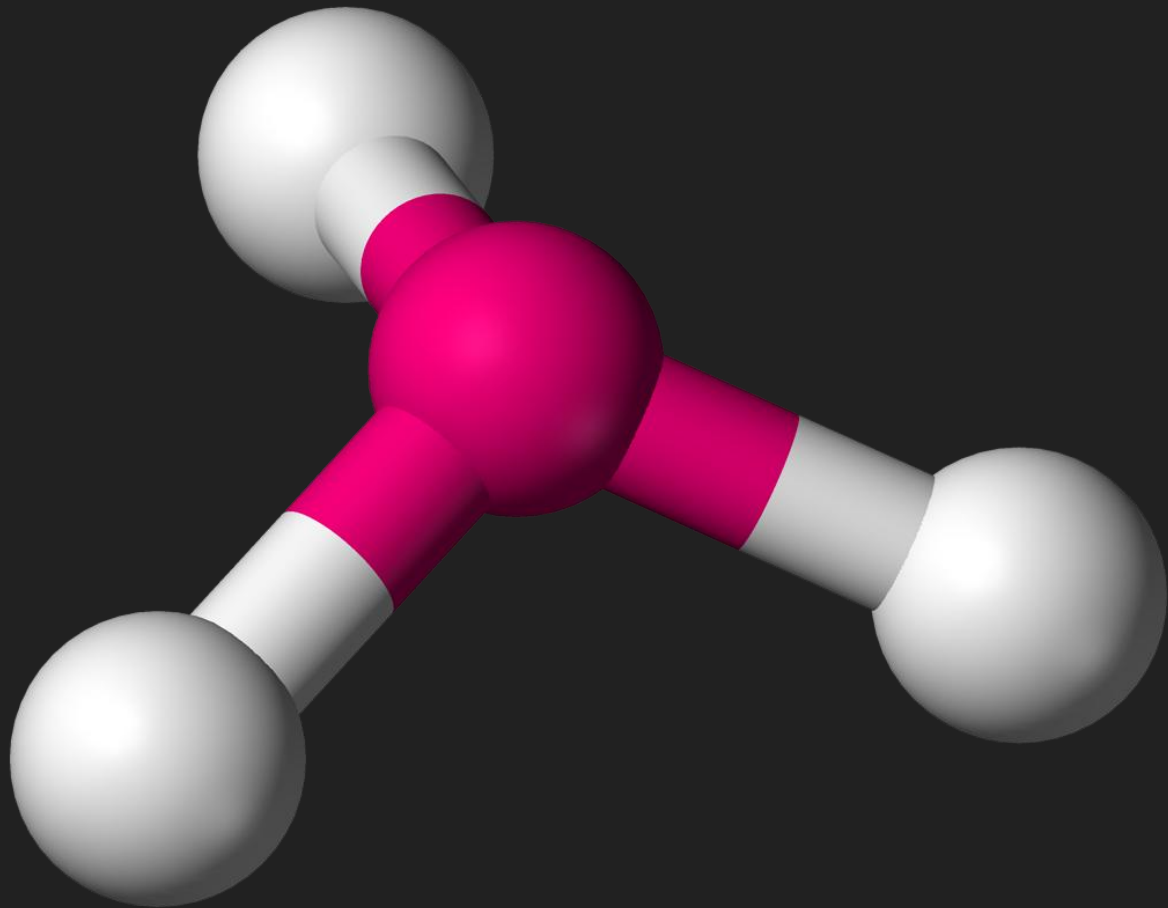
4bp, 1lp

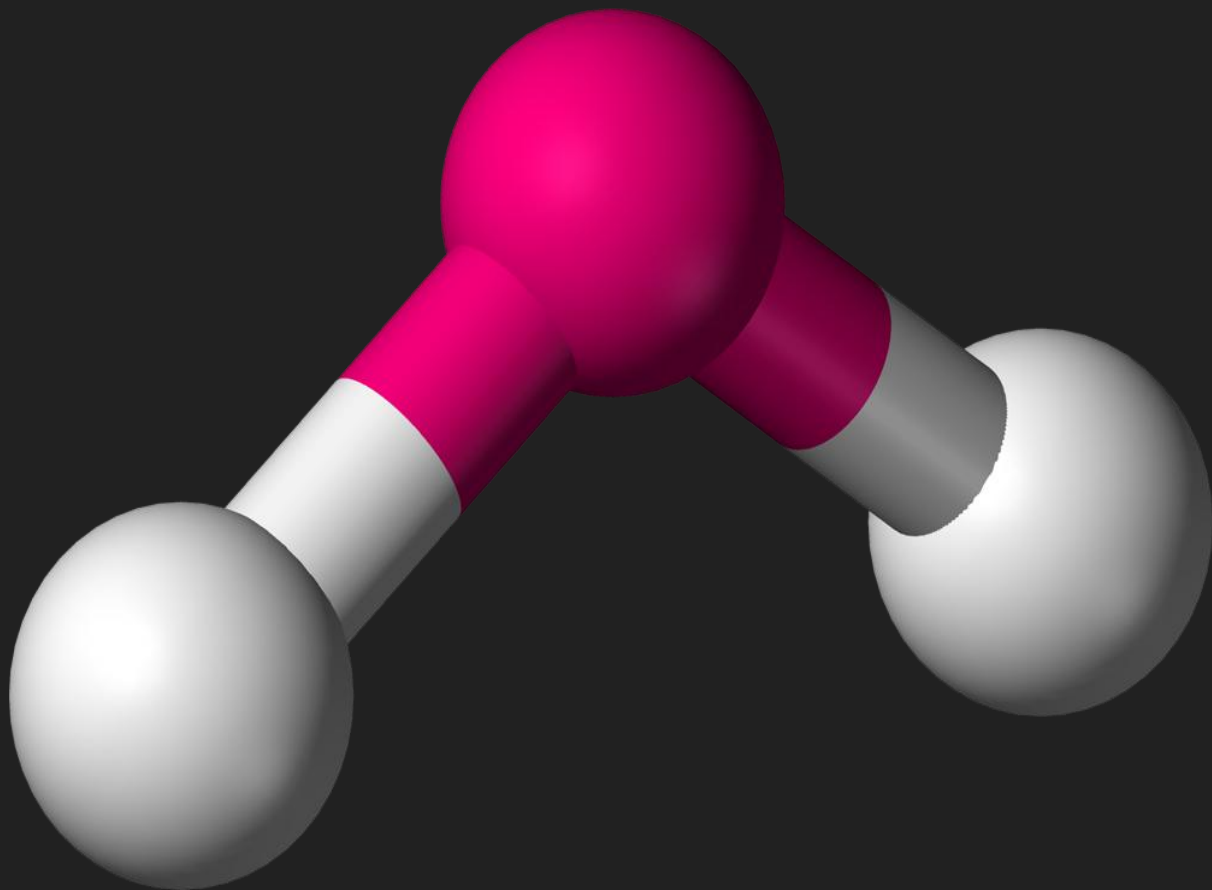
See saw



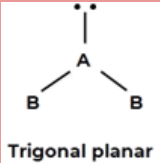
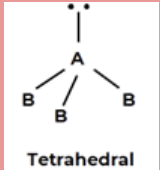
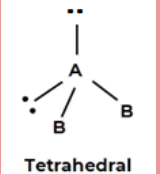
3bp, 2lp

T-shape

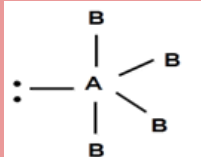


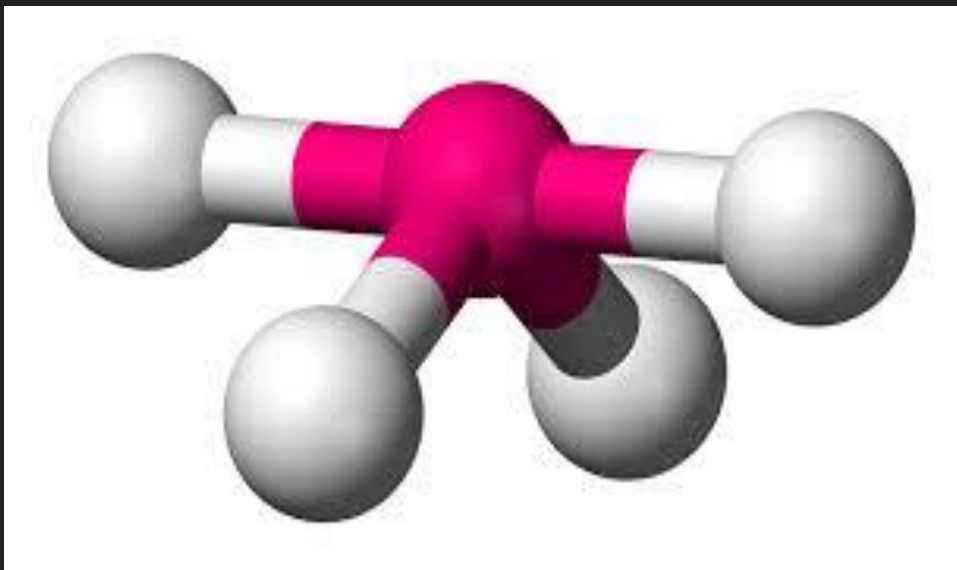


Geometry in which central atom has one or more lp of e⁻

Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_2E	2	1	 Trigonal planar	Bent, SO_2 , O_3
AB_3E	3	1	 Tetrahedral	Trigonal Pyramidal, NH_3
AB_2E_2	2	2	 Tetrahedral	Bent, H_2O

Geometry in which central atom has one or more lp of e⁻

Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_4E	4	1	 Trigonal bi-pyramidal	See saw, SF_4

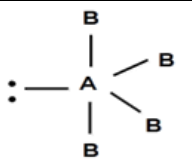
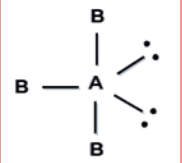


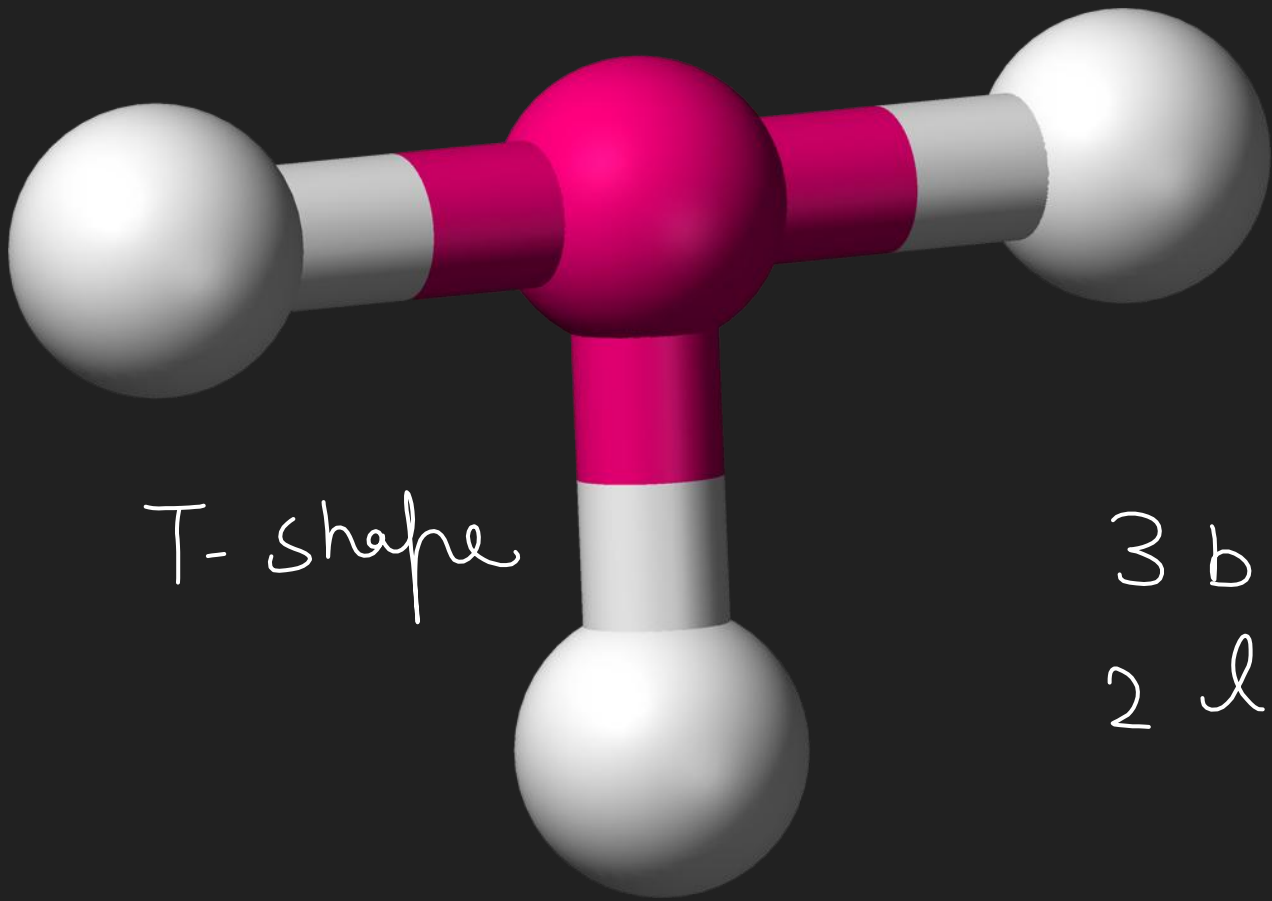
See Saw

4 b.p

1 l.p

Geometry in which central atom has one or more lp of e⁻

Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_4E	4	1	 Trigonal bi-pyramidal	See saw, SF_4
AB_3E_2	3	2	 Trigonal bi-pyramidal	T- Shape, ClF_3

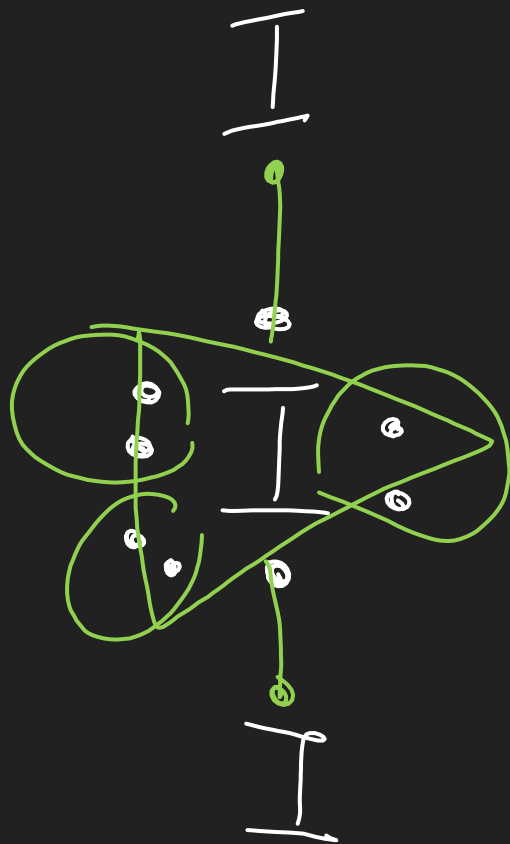


T-shape

3 b p
2 l p

P4Q

I_3^-

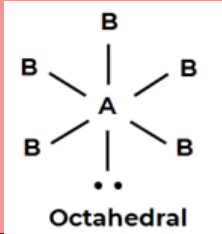


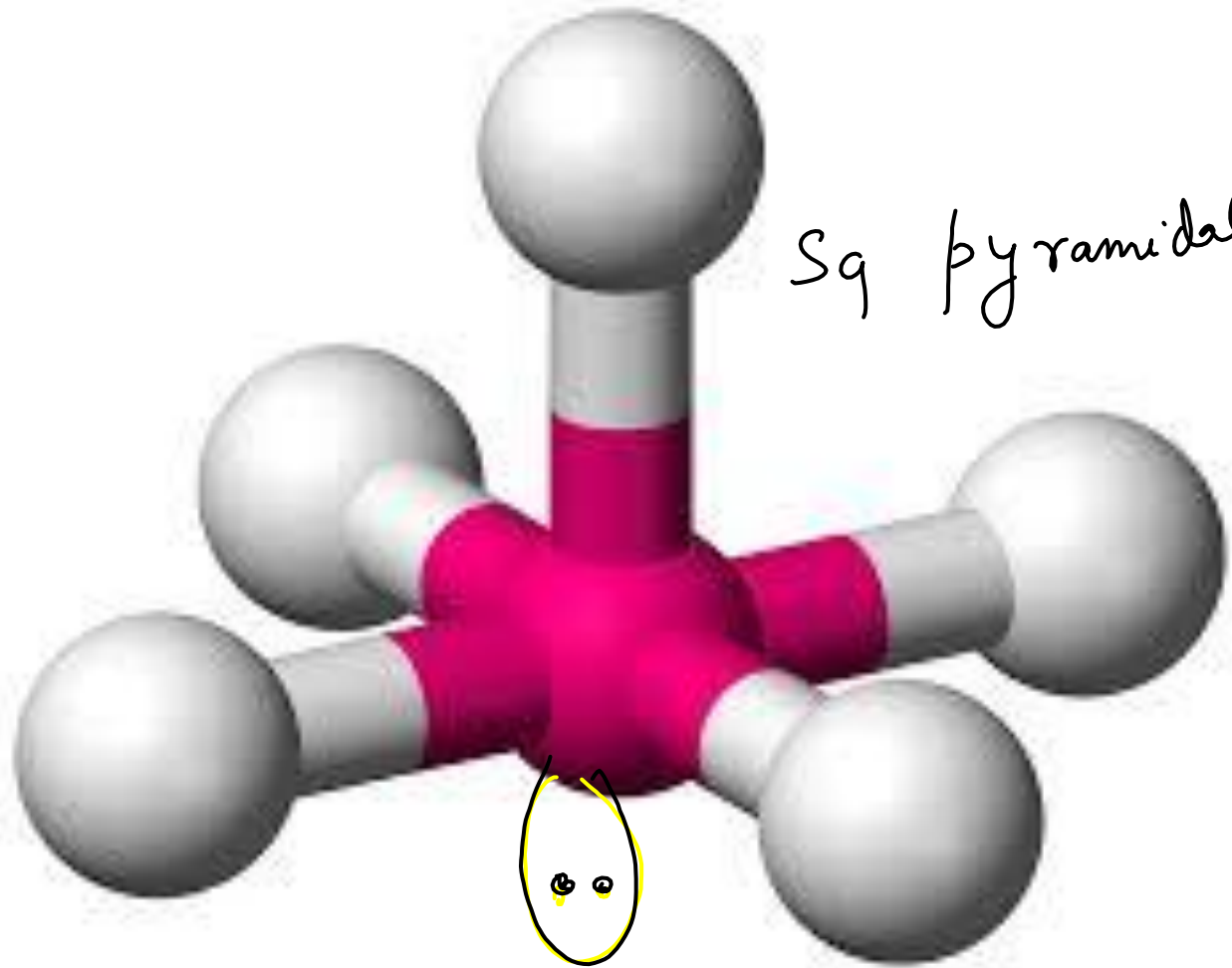
3 l p.

2 b p

Linear

Geometry in which central atom has one or more lp of e⁻

Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_5E	5	1	 Octahedral	Square Pyramidal, BrF_5

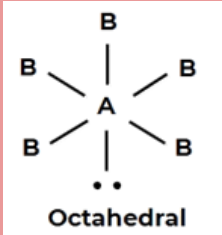


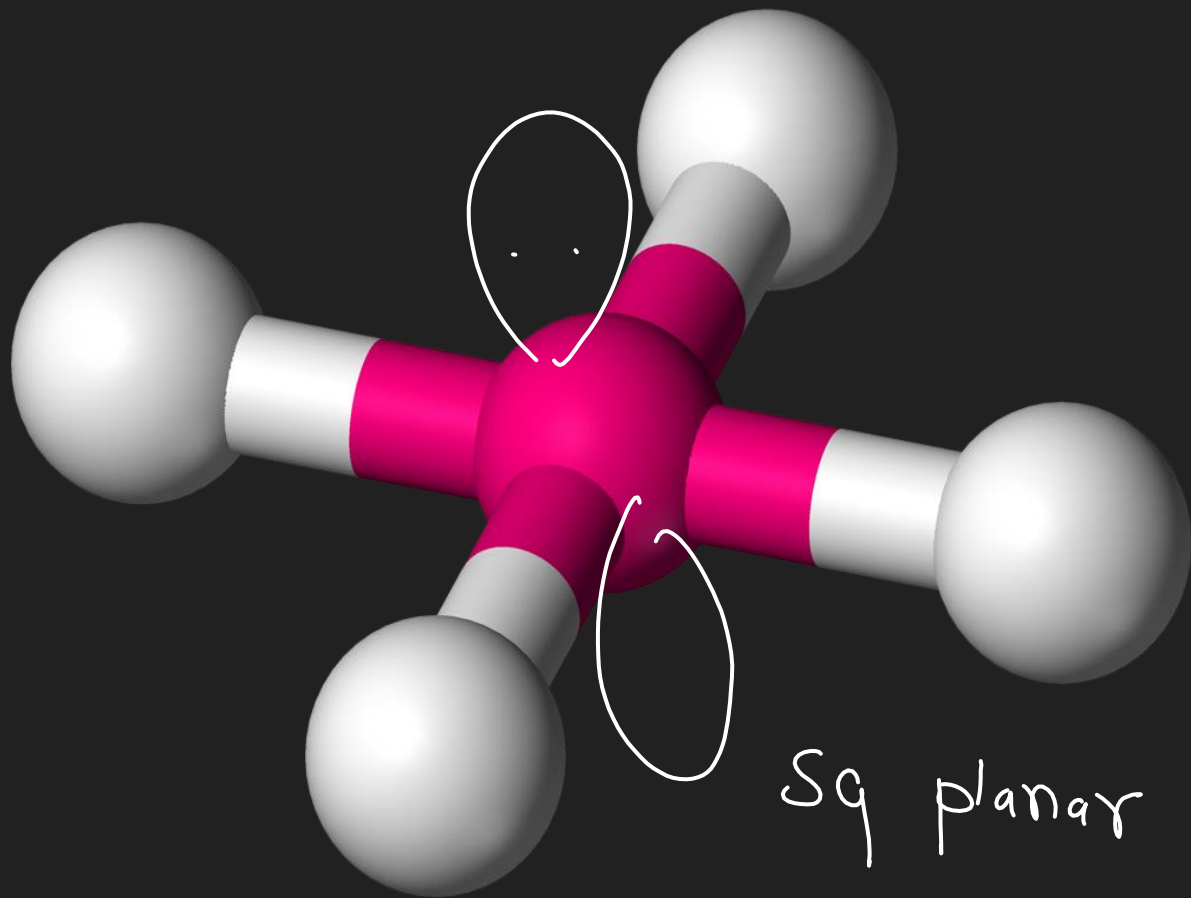
Sq pyramidal

5 b.p.

1 l.p.

Geometry in which central atom has one or more lp of e⁻

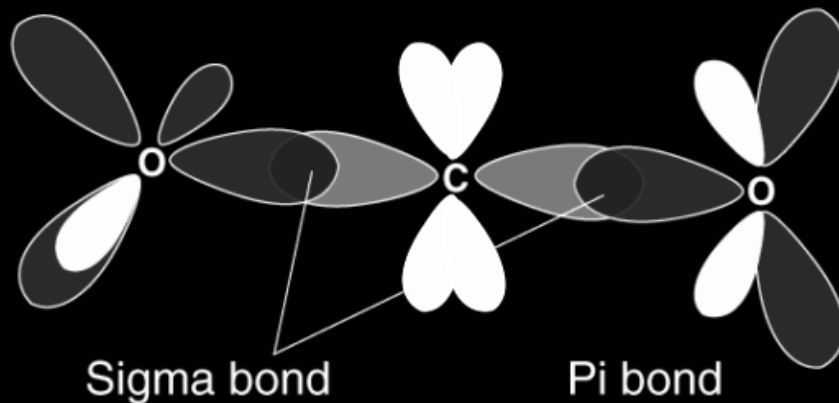
Molecule Type	No. of Bonding Pairs	No. of lone pairs	Arrangement of electron pairs	Shapes and Examples
AB_5E	5	1		Square Pyramidal, BrF_5
AB_4E_2	4	2		Square Pyramidal, XeF_4



Sq planar

26 p.
41 p

Valence Bond theory





**Walter
Heitler**



Fritz W. London

Presented by Heitler & London to explain how a covalent bond is formed.

Valence Bond Theory

- ➡ To form a covalent bond overlapping occurs between half filled valence shell orbitals of the two atoms
- ➡ Resulting bond acquires a pair of electrons with opposite spins to get stability
- ➡ Orbitals come closer to each other from the direction in which there is maximum overlapping
- ➡ So covalent bond has directional character.
- ➡ Extent of overlapping $\propto$ **strength of chemical bond.**

Valence Bond Theory

👉 Extent of overlapping depend on two factors.

❑ Nature of orbitals

p, d and f are directional orbitals → more overlapping

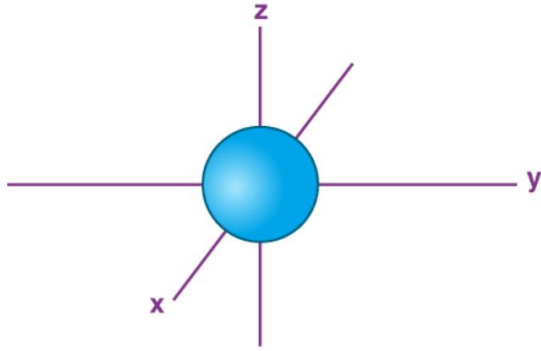
s-orbitals → non directional - less overlapping

❑ Nature of overlapping

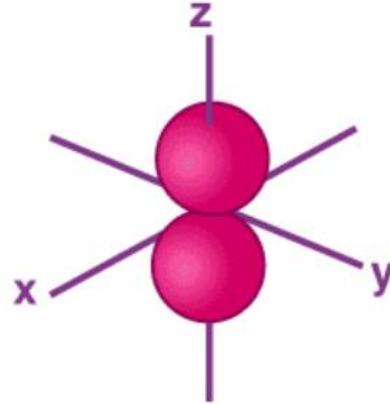
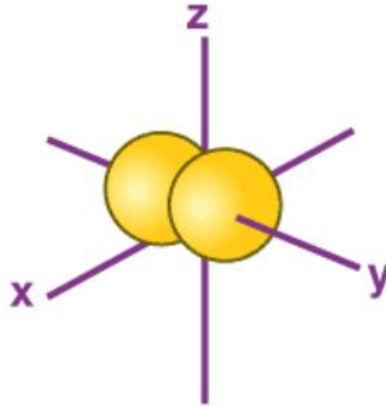
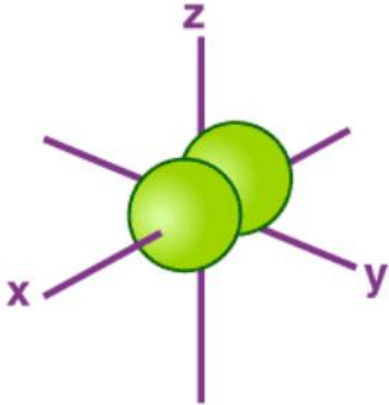
Co-axial overlapping - extent of overlapping more

Collateral overlapping - extent of overlapping less

Shape of orbitals

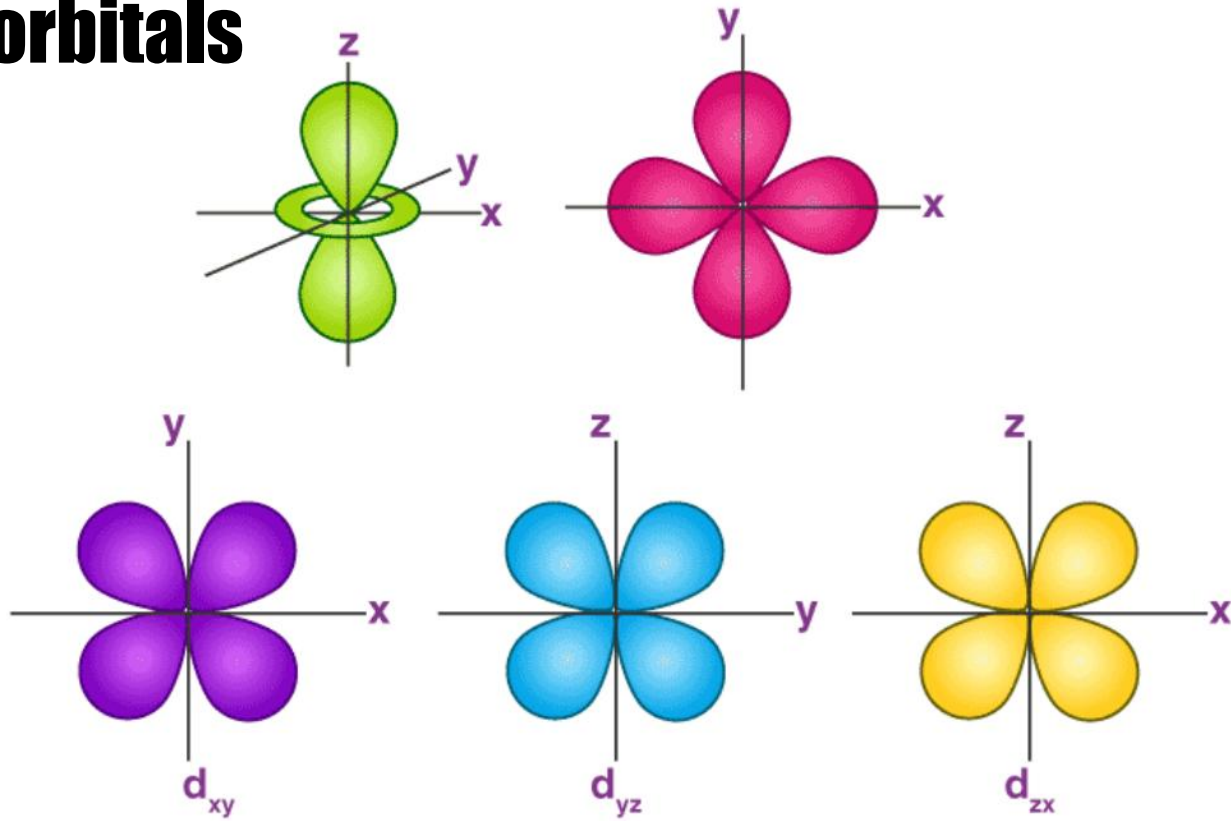


s-orbital



p-orbitals

Shape of orbitals



d-orbitals

Type of Overlap & Bond

Covalent Bond

σ -bond

Head on overlap/
axial overlap



π -bond

Lateral/Parallel/
sideway overlap

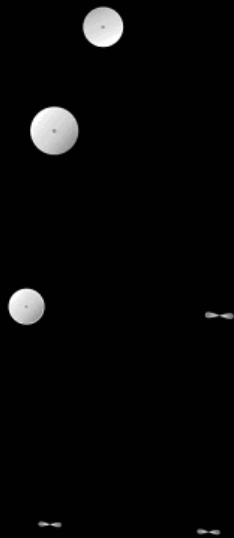
π Bond



π Bond



σ -bond



π -bond

π Bond



π Bond



Sigma bonds



Sigma Bond

Formed by end to end overlap (head-on) of bonding orbitals along the Internuclear axis

Also called head on overlap or axial overlap

Combinations of orbitals

```
graph TD; A[Combinations of orbitals] --> B[s-s overlapping]; A --> C[s-p overlapping]; A --> D[p-p overlapping]
```

s-s overlapping

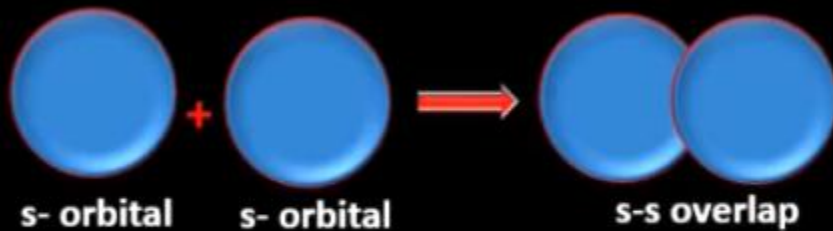
s-p overlapping

p-p overlapping

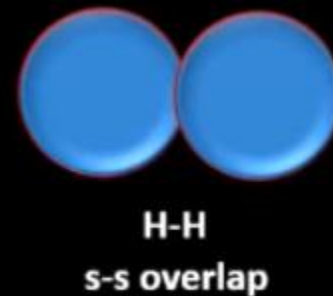
Order of strength of Co-axial overlapping -

p - p > s - p > s - s

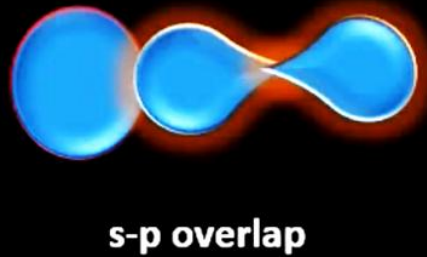
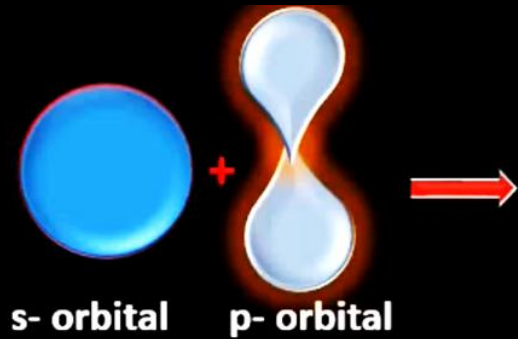
s-s overlapping



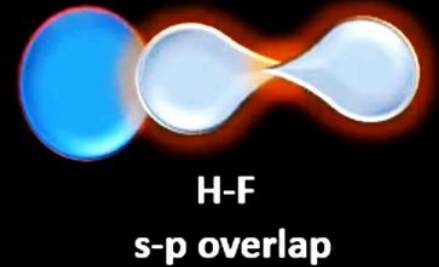
Example



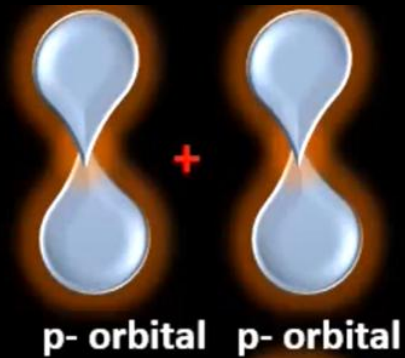
***s-p** overlapping*



Example



p-p overlapping



Example



pi bonds

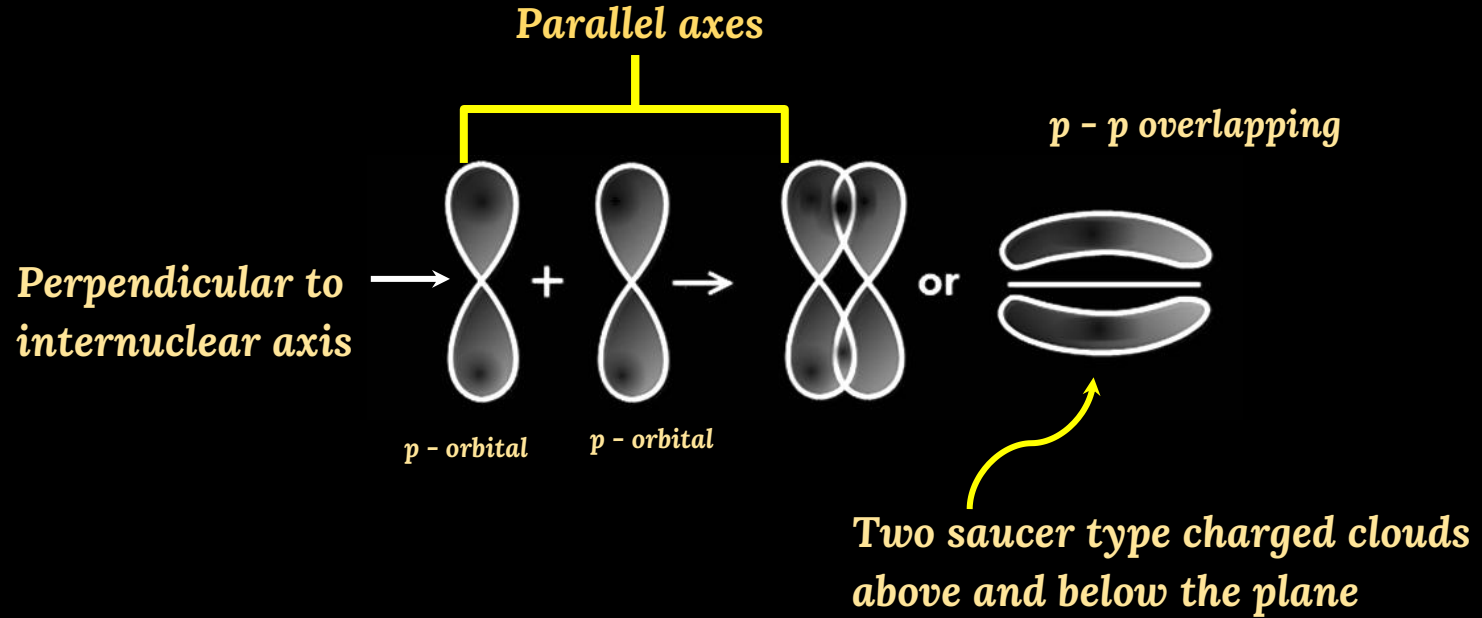


Pi (π) bond :

- *This type of covalent bond is formed by the side wise (or lateral) overlapping of half filled atomic orbitals of bonding atoms.*

pi bonds

.....



... Difference between sigma and pi bond ...

<i>Sigma (σ)</i>	<i>Pi (π)</i>
<i>1. End – to – end overlap</i>	<i>1. Parallel overlap</i>
<i>2. Single e^- cloud present symmetrically about the internuclear axis</i>	<i>2. Two e^- cloud present above and below the internuclear axis</i>
<i>3. Stronger bond</i> <i>Ex : C – C 80 Kcal / mol</i>	<i>3. Weaker bond</i> <i>Ex : C – C 65 Kcal / mol</i>
<i>4. More stable bond</i>	<i>4. Less stable bond</i>
<i>5. Can independently exist</i>	<i>5. π bond is formed after the σ bond</i>

... Difference between sigma and pi bond ...

	Sigma (σ)		Pi (π)
6.	<i>Free rotation about σ bonds is possible about the internuclear axis</i>	6.	<i>Two e^- cloud present above and below the internuclear axis So free rotation is not possible</i>
7.	<i>σ bonds e^- are localized</i>	7.	<i>π bond e^- are delocalized</i>
8.	<i>Shape of the molecule is determined by the σ bonds</i>	8.	<i>Shape is not affected by the number of π bonds</i>

Among the following which forms a bond?

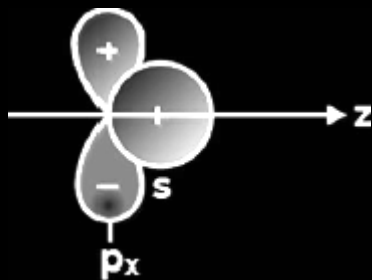
A



B



C



D



Among the following which forms a bond?

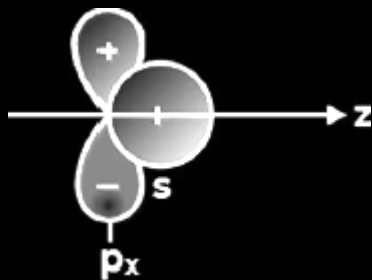
A



B



C

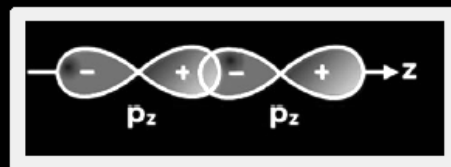
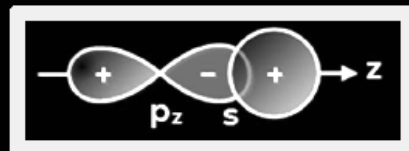


D

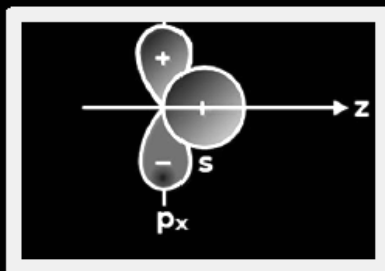


Solution :

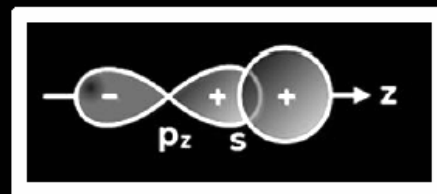
Negative overlap



Zero overlap



Positive overlap



The s-orbital does not show preference to any direction because_____

A

It is the smallest orbital

B

It is present in every atom

C

It is spherically symmetric

D

It is the first orbital

The s-orbital does not show preference to any direction because_____

A

It is the smallest orbital

B

It is present in every atom

C

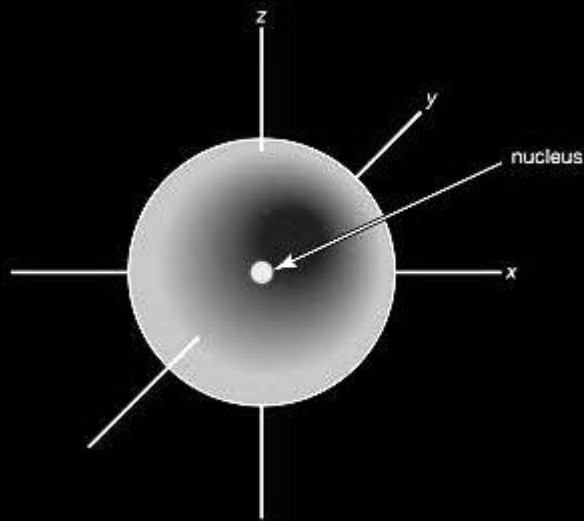
It is spherically symmetric

D

It is the first orbital

Solution :

Since it is spherical it is same in all the directions.

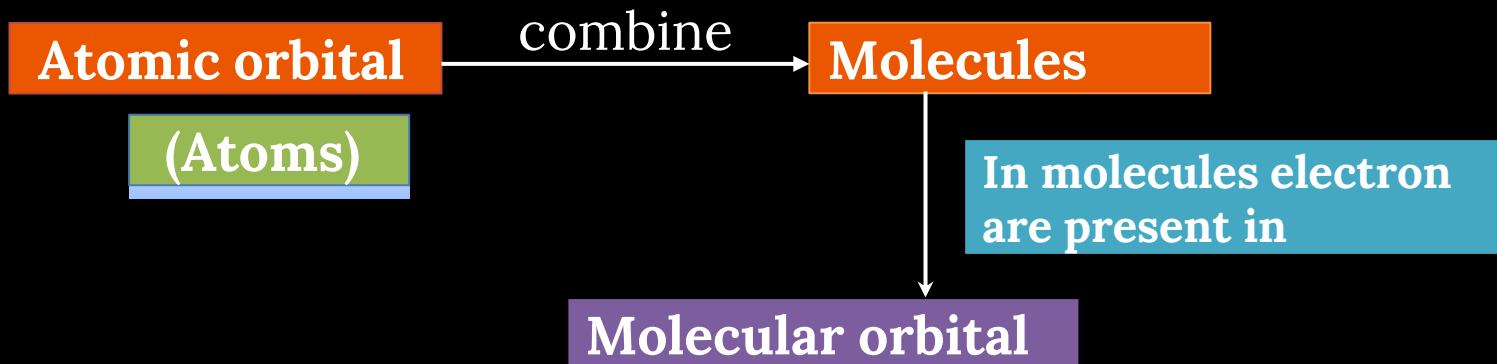


MOLECULAR ORBITAL THEORY(MOT)

Introduced by F.Hund and R.S. Mulliken

VBT considered O_2 as diamagnetic. Experimentally O_2 was found to be paramagnetic. Magnetic behavior of this molecule was explained by this theory.

Molecular Orbital Theory



What are the types of Atomic Orbitals??

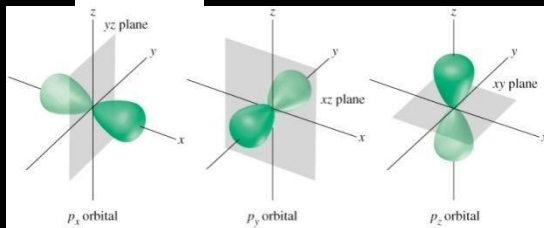
Type of Atomic Orbitals

S Orbital



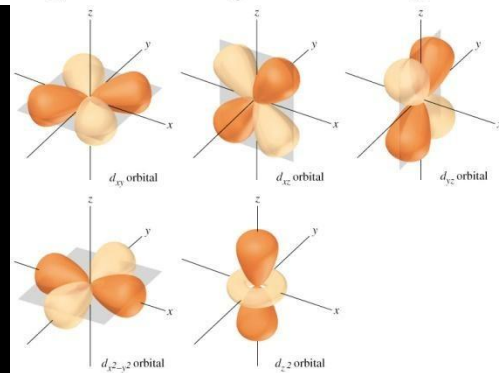
Spherical in shape

p orbital



Double in shape

d Orbital

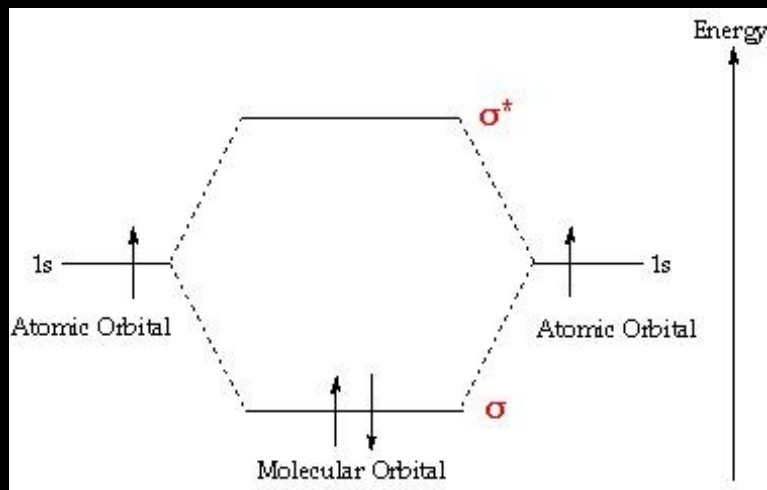


Double Dumbbell in shape

Postulates of Molecular Orbital Theory (MOT)

POSTULATES OF MOT

1. The electrons in molecules are present in various molecular orbital as the electron of atom are present in various atomic orbital.



POSTULATES OF MOT

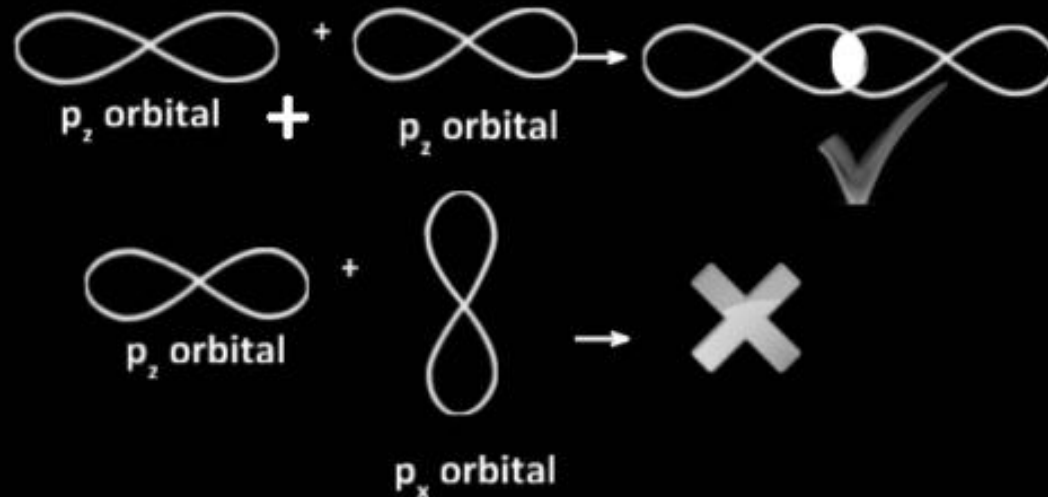
2. The atomic orbitals of comparable energies and proper symmetry combine to form molecular orbital.



- This means that 1s can combine with only 1s.
- 1s cannot combine with 2s.
- Reason : 2s has higher energy.

POSTULATES OF MOT

2. The atomic orbitals of comparable energies and proper symmetry combine to form molecular orbital.



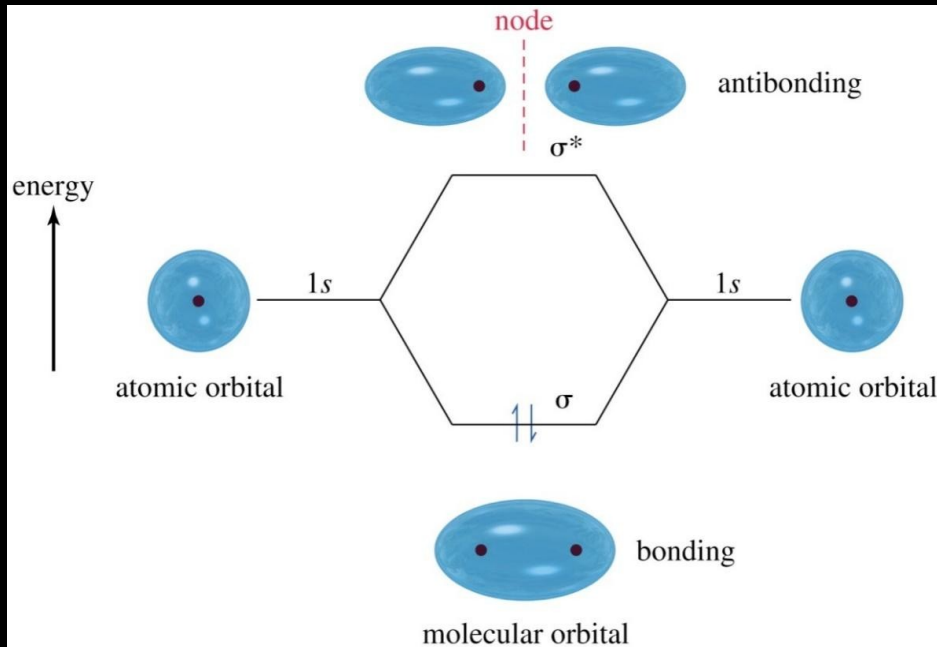
POSTULATES OF MOT

3. An electron in an atomic orbital is influenced by one nucleus, but in molecular orbital it is influenced by two or more nuclei.

So, atomic orbital was monocentric but Molecular orbitals are polycentric

POSTULATES OF MOT

4) The number of molecular orbital formed is equal to the number of combining atomic orbitals. When two atomic orbitals combine, two molecular orbitals are formed. One is known as bonding molecular orbital while the other is called antibonding molecular orbital.



POSTULATES OF MOT

5) The bonding molecular orbital has lower energy and hence greater stability than the corresponding antibonding molecular orbital.

POSTULATES OF MOT

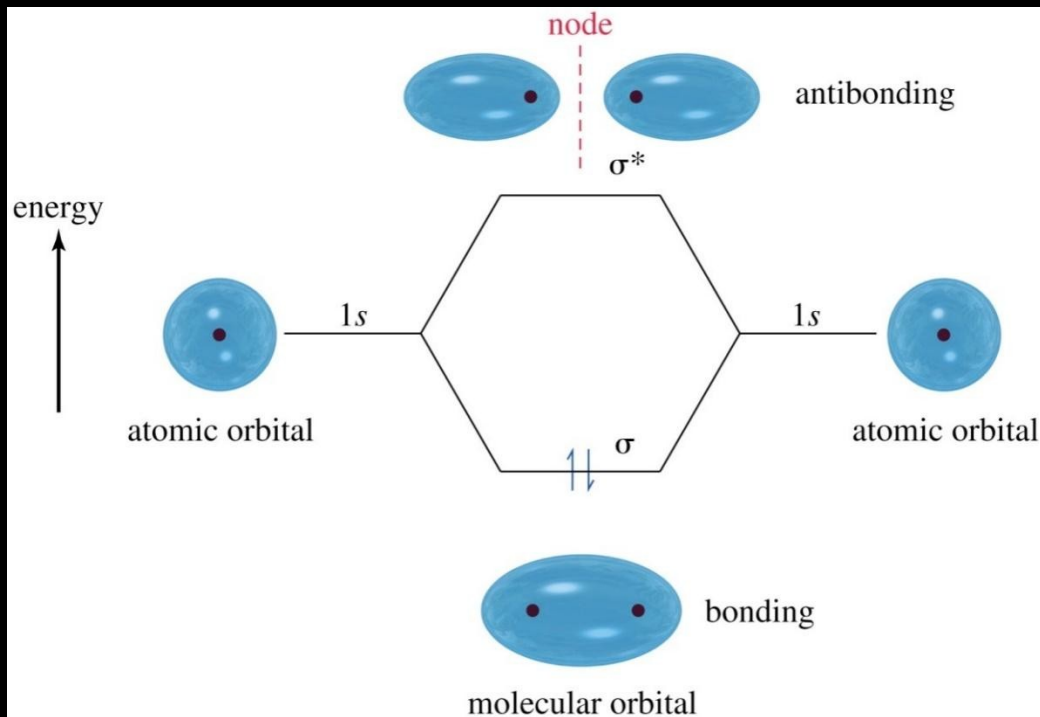
6) Just as the electron probability distribution around a nucleus in an atom is given by an atomic orbital, the electron probability distribution around a group of nuclei in a molecule is given by a molecular orbital.

POSTULATES OF MOT

7) The molecular orbitals like atomic orbitals are filled in accordance with the Aufbau principle, obeying the Pauli's exclusion principle and the Hund's rule.

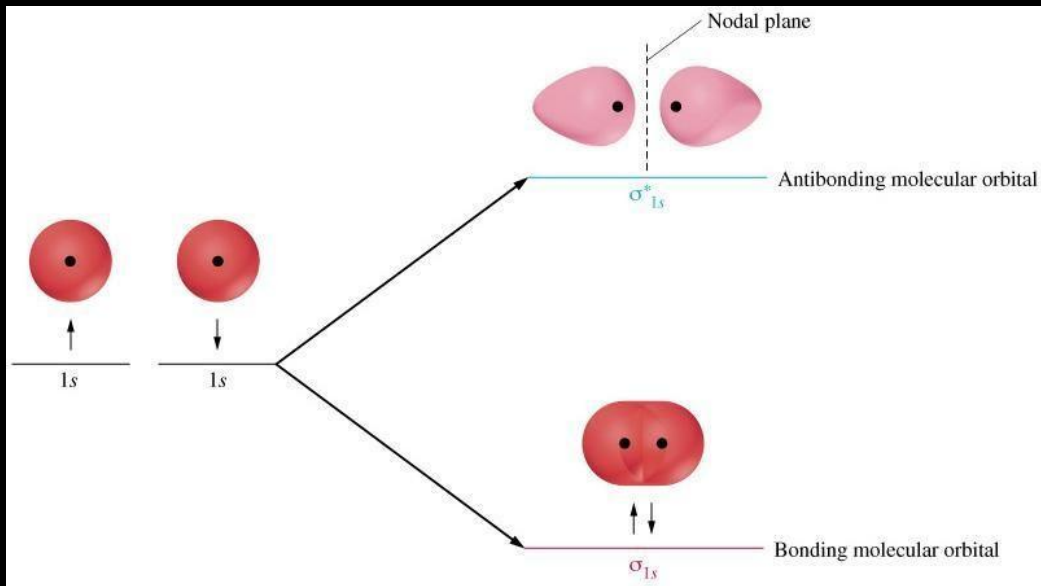
Types M.O. and Energy Levels

Energy of Molecular Orbital



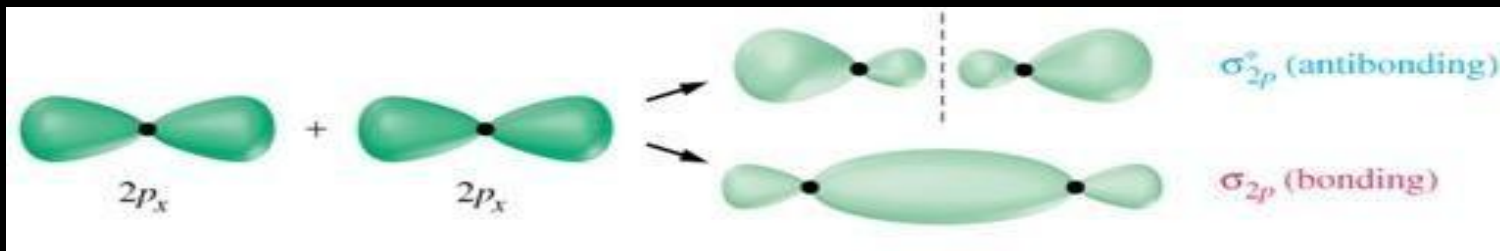
**Note:- S Orbitals forming sigma bond and
P Orbitals forming sigma and pi bond**

S Orbitals Forming sigma bond

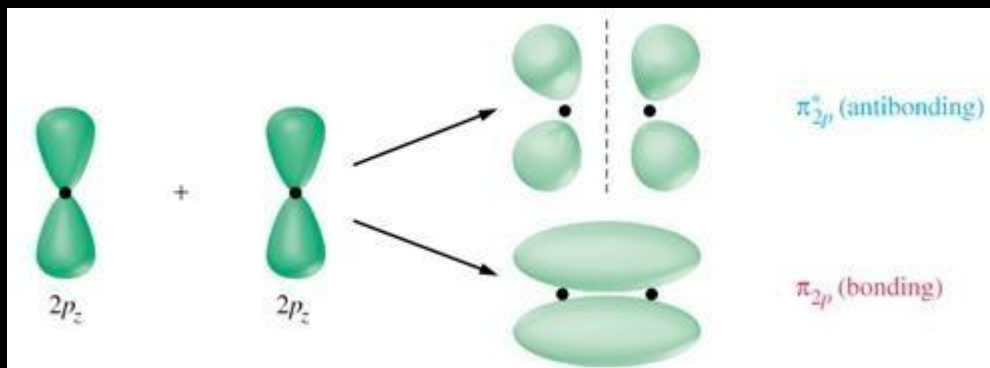


p orbital
p orbital
x

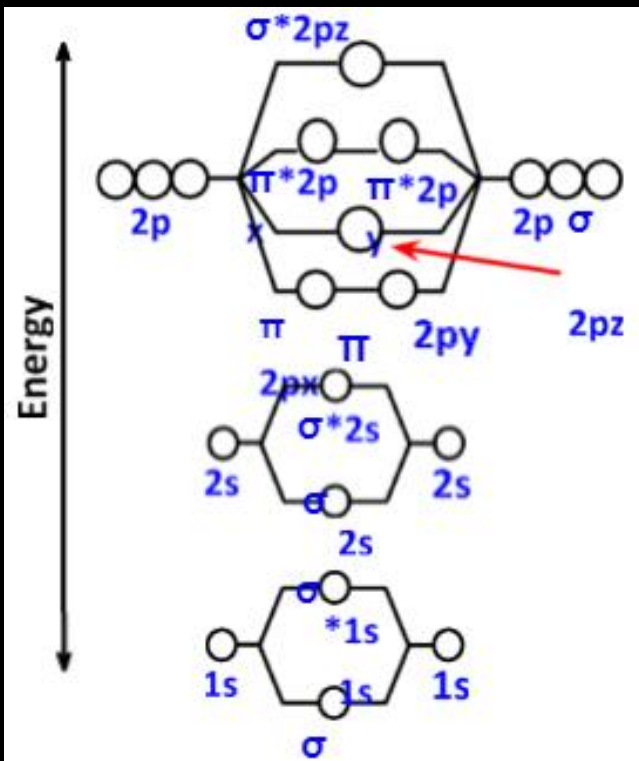
P Orbitals Forming Sigma bond



P Orbitals Forming Pi bond

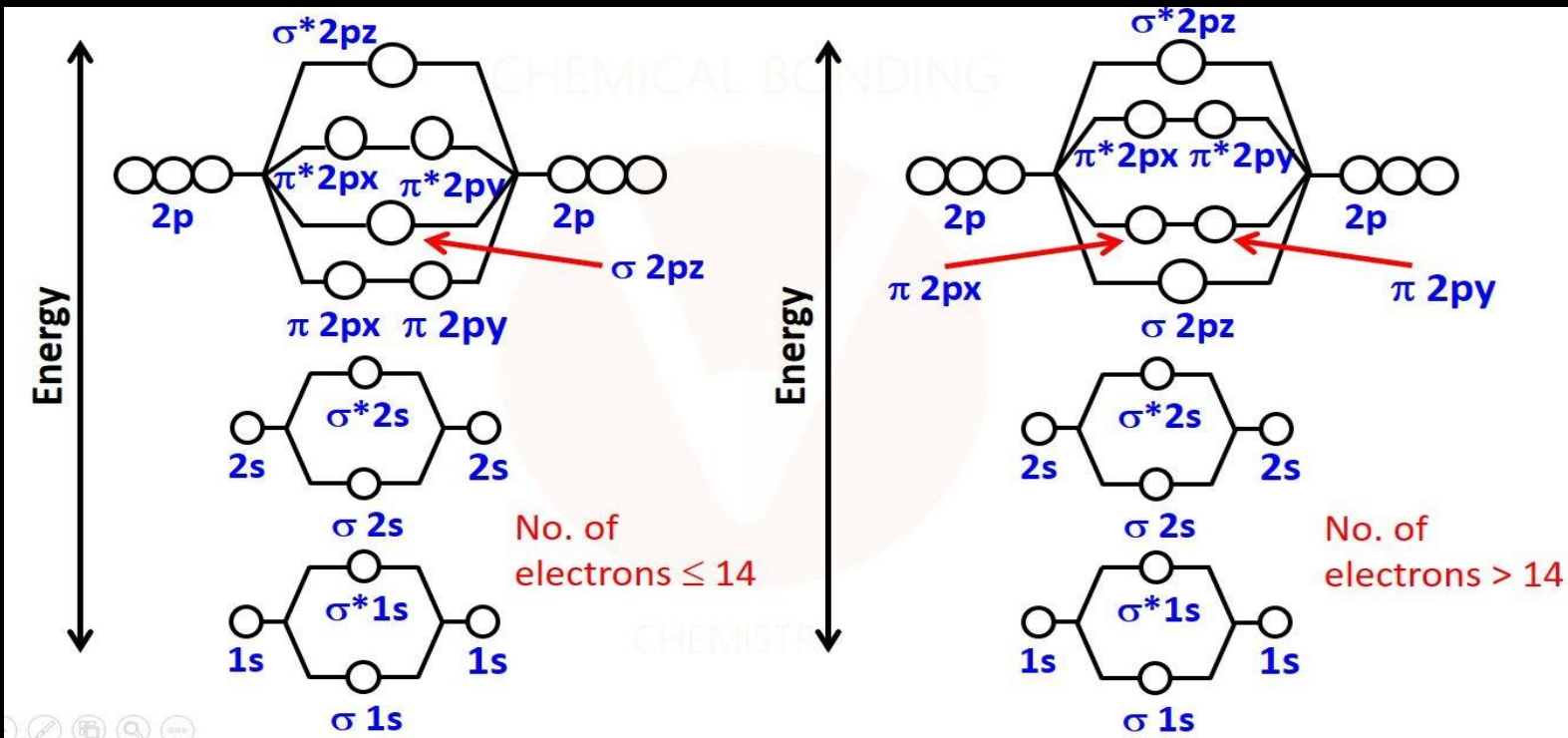


MOLECULAR ORBITAL DIAGRAM



Energy of Orbitals when total
Electrons in a molecule are ≤ 14

MOLECULAR ORBITAL DIAGRAM



What information can be obtained from the MOT Theory

□ **Bond Order**

□

**Stability of
Molecules**

□

Nature of Bond

□

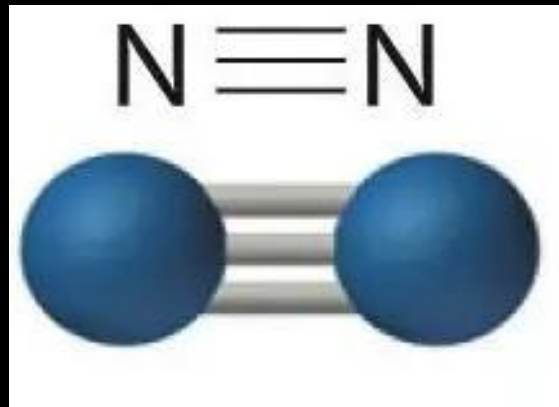
Magnetic Nature

□

Bond Length

Bond Order

Bond order is the number of chemical **bonds** between a pair of atoms.



Bond order

$$\text{Bond order (B.O.)} = \frac{1}{2} (N_b - N_a)$$

N_b = number of electrons in bonding orbitals
 N_a = number of electrons in antibonding orbitals

Nature of the bond

If Bond Order = 1, Single
Covalent Bond
e.g. H_2 , F_2 Molecule

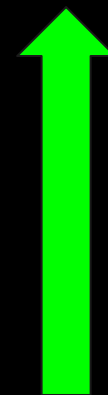
If Bond Order = 2, Double
Covalent Bond
e.g. O_2 Molecule

If Bond Order = 3, Triple
Covalent Bond
e.g. N_2 Molecule

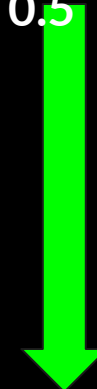
Trick



No. of e-	Bond Order
9	0.5
10	1
11	1.5
12	2
13	2.5
14	3
15	2.5
16	2
17	1.5
18	1
19	0.5



-
0.5



Example



Molecules	Total e-	Bond Order
CO		
NO		
NO ⁺		
O ₂		
O ₂ ⁺		
O ₂ ⁻²		
O ₂ ⁻¹		

Stability Of Molecules

If N_b = number of electrons in bonding orbitals
and N_a = number of electrons in antibonding orbitals

Case 1 :
If $N_b > N_a$
Molecule is stable

More bonding orbitals are occupied. Hence a stable molecule is formed.

e.g. O_2 , N_2

Stability Of Molecules

Case 2 :
If $N_b > N_a$
Molecule is unstable

Antibonding influence is stronger Hence an unstable molecule is formed.

Stability Of Molecules

Case 3:
If $N_b = N_a$ Molecule is
Unstable

Influence of antibonding Electrons is slightly more Than bonding electrons

Bond Length

Bond length is inversely proportional to bond order

More is the bond order, lesser is the bond length

Magnetic

If all the molecular orbitals in a molecule are doubly occupied, the substance is DIAMAGNETIC.

**If one or more molecular orbitals are singly occupied, the substance is
PARAMAGNETIC.**

Super Short Trick :

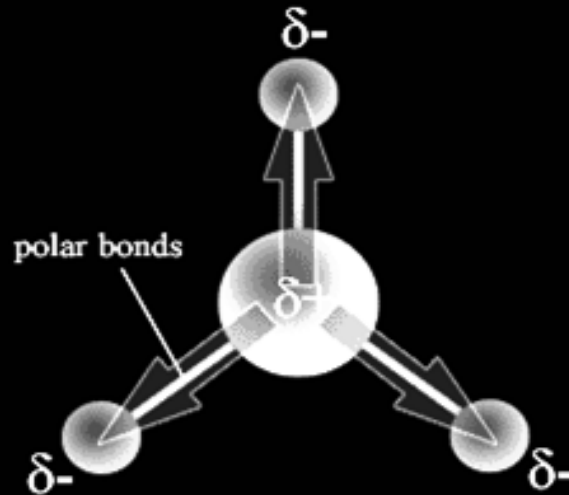
Paramagnetic :-

When total
no. of e- in
Diatomic Specie
are Odd,10,16.

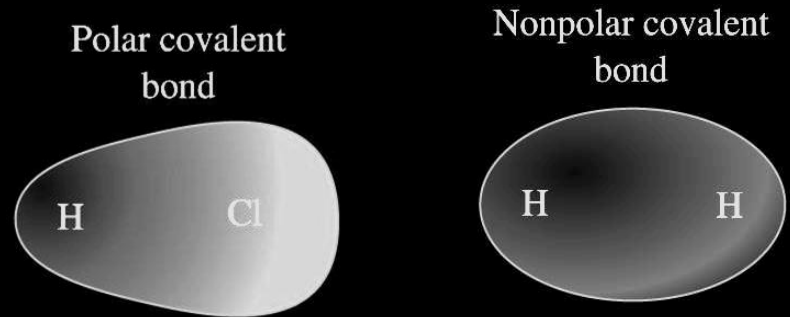
Diamagnetic

When Total no.
of e- are Even

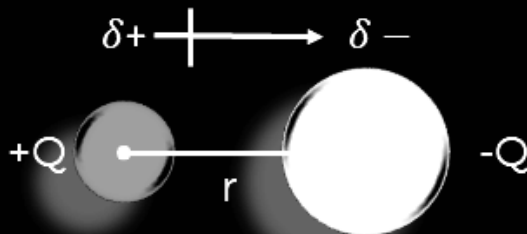
Dipole moment



- Dipole moment is represented by the crossed arrow ($\text{---} \text{+} \text{---} \text{---} \text{---}$) put on lewis structure of the molecule.



Dipole moment



$$\text{Dipole moment } (\mu) = \text{charge}(Q) \times \text{distance of separation } (r)$$

In coulombs

In meters

Usually
expressed
in Debye
units

$$1D = 3.33564 \times 10^{-30} \text{ Cm}$$

Dipole moment

- In the diatomic molecule μ depends upon difference of electronegativity

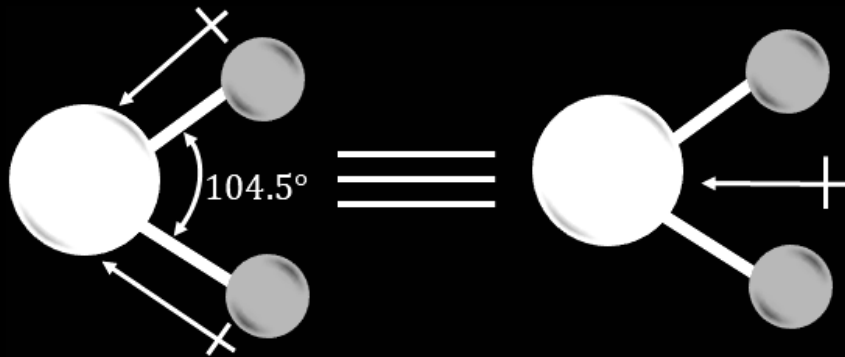
$$\mu \propto EN$$

Order of μ : $\text{H-F} > \text{H-Cl} > \text{H-Br} > \text{H-I}$

$\mu = 0$ for $\text{H-H}, \text{F-F}, \text{Cl-Cl}, \text{Br-Br}, \text{O-O}$

Dipole moment

For **polyatomic molecules** μ depends on the vector sum of dipole moments of all the covalent bonds.



μ of a molecule is the vector sum of the μ of various bonds

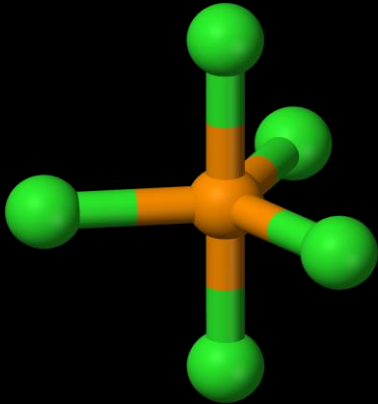
Application of Dipole moment

- To determine polarity and geometry of molecule -
- To calculate % ionic character :-

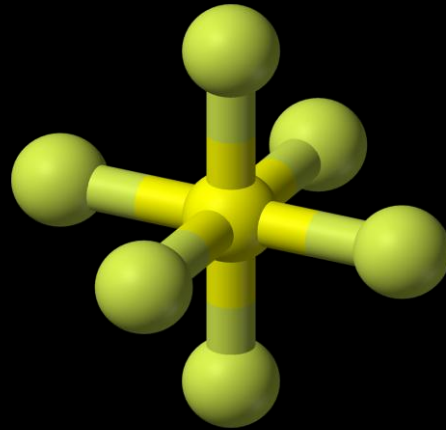
$$\% \text{ Ionic character} = \frac{\text{Experimental value of } \mu}{\text{Theoretical value of } \mu} \times 100$$

Dipole moment

- For PCl_5 and SF_6 , etc. $\mu = 0$ due to their regular geometry.



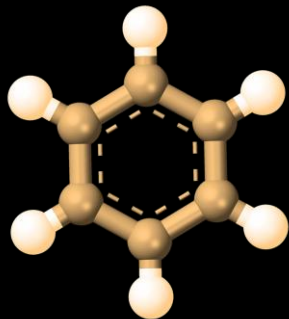
PCl₅



SF₆

Dipole moment

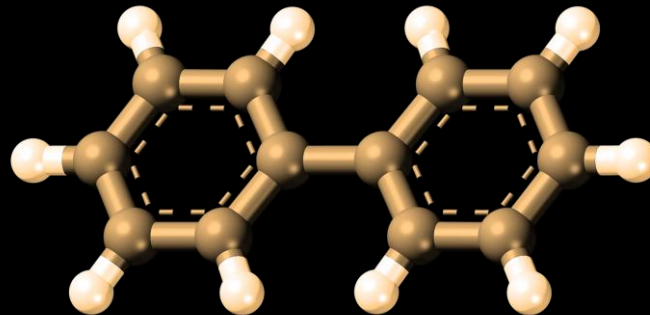
- Benzene, naphthalene, biphenyl have $\mu = 0$ due to planar structure.



Benzene

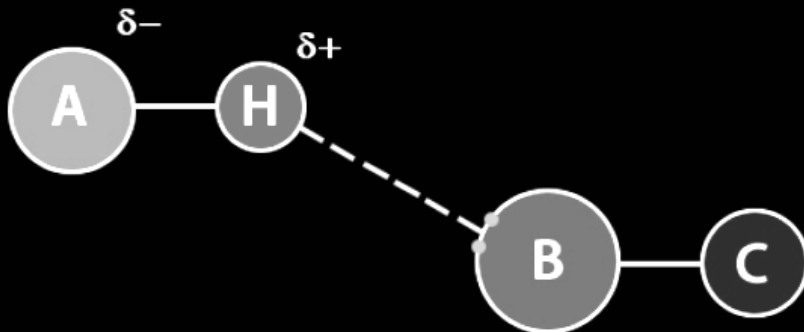


Naphthalene

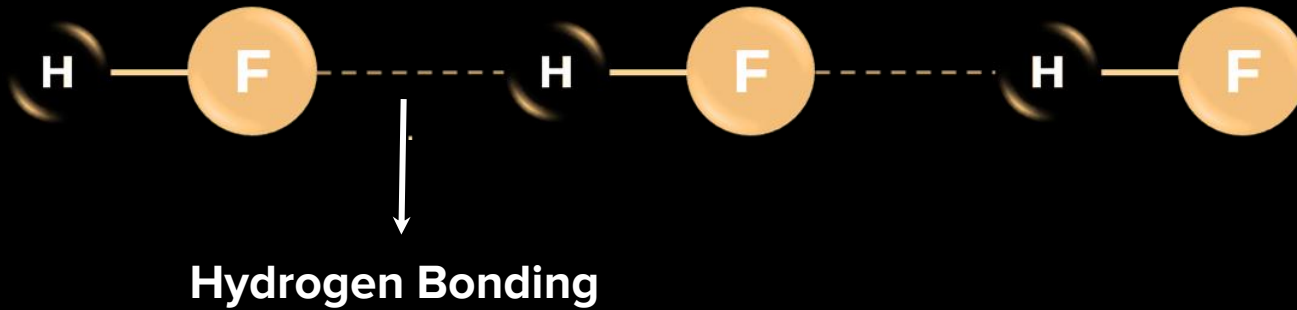


Biphenyl

Hydrogen Bonding

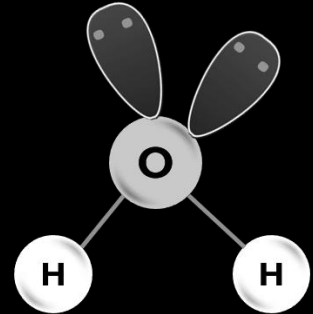
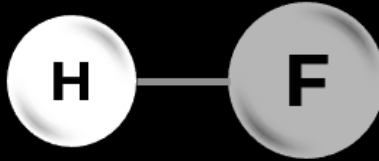
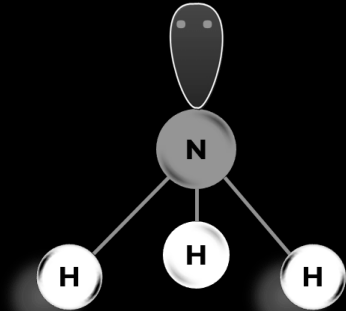


Hydrogen Bonding



Hydrogen Bonding

It is an electrostatic attractive force between covalently bonded H atom of one molecule and an electronegative atom (F, O, N)



Hydrogen Bonding



Condition for Hydrogen Bonding

```
graph TD; A[Condition for Hydrogen Bonding] --> B[Hydrogen Should be Bonded with high electronegative element like, F, O, N]; A --> C[Atomic size of electronegative element should be small];
```

Hydrogen Should be Bonded with high electronegative element like, F, O, N

Atomic size of electronegative element should be small

Types of Hydrogen Bonding

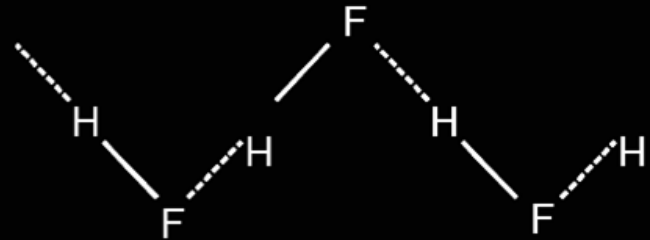
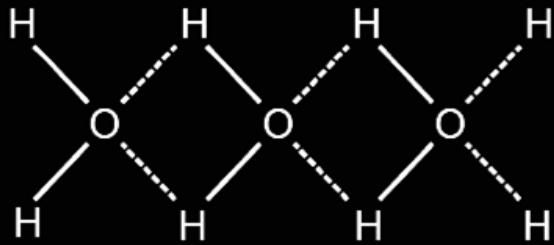
```
graph TD; A[Types of Hydrogen Bonding] --> B[Intermolecular]; A --> C[Intramolecular]
```

Intermolecular

Intramolecular

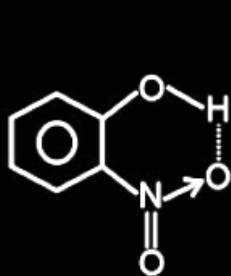
Intermolecular Hydrogen bond

H-bond formation between two or more molecules of either the same or different compounds known as Intermolecular H-bonding.

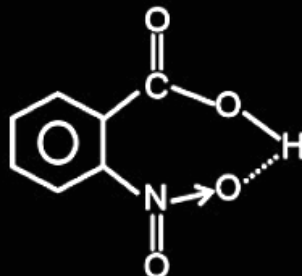


Intramolecular Hydrogen bond

- ❖ It takes place within the molecule.
- ❖ H-bonded with EN elements of a functional group, form H-bond with another EN element present on nearest position on the same molecule.
- ❖ This type of H-bond is mostly occurred in organic compounds.



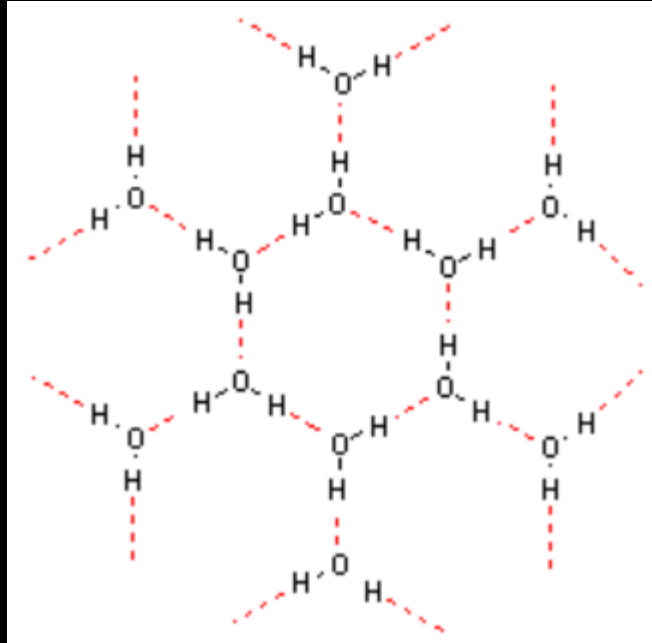
ortho nitrophenol



ortho benzoic acid

Applications of Hydrogen Bonding

Applications of Hydrogen Bonding

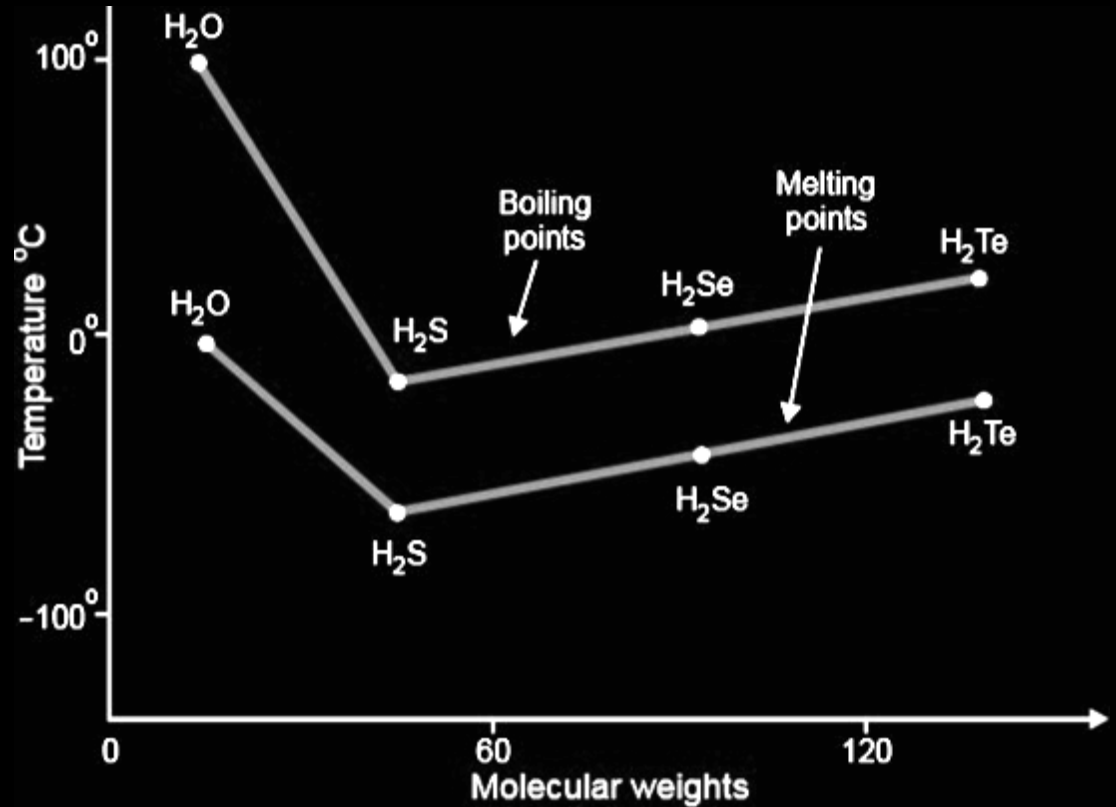


Effect of hydrogen bond

Boiling point and Melting Point:

- *Due to Intermolecular Hydrogen bonding, Melting Point & Boiling Point of compounds increases.*

Boiling point and Melting Point:



Effect of hydrogen bond

Solubility :

Effect of hydrogen bond

Dimerisation of acids :

Effect of hydrogen bond

Viscosity:

- *Hydrogen bond associates molecules together, so viscosity increases*

LET'S SOLVE QUESTIONS.....

Previous year questions

Chemical Bonding
NSEJS

Hydrogen fluoride is a liquid at room temperature due to :

NSEJS 2012

A

Dimerisation

B

Dissociation followed by aggregation

C

Association

D

polymerisation

Hydrogen fluoride is a liquid at room temperature due to :

NSEJS 2012

A

Dimerisation

B

Dissociation followed by aggregation

C

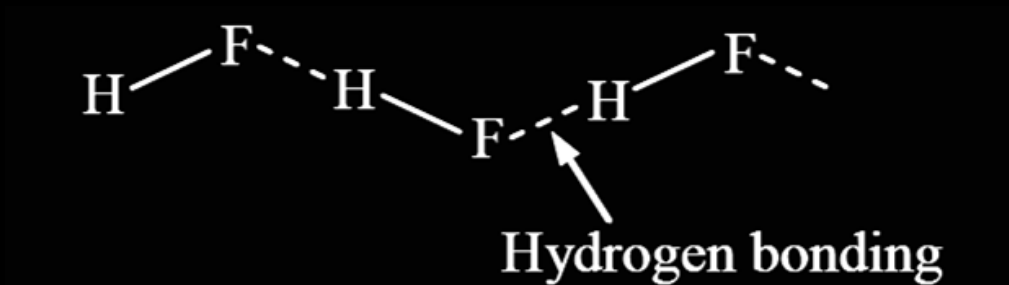
Association

D

polymerisation

Solution :

Association of hydrogen bonding.



The Lewis acid strength of BBr_3 , BCl_3 and BF_3 is in the order

A



B



C



D



The Lewis acid strength of BBr_3 , BCl_3 and BF_3 is in the order

A



B



C



D



Solution:

Order of back bonding BF_3 BCl_3 $>$
Back bonding $2p - 2p$ $2p - 3p$ $2p - 4p$ Stronger is the back bonding
weaker the tendency to act as a Lewis acid Br_3



The H-C-H , H-N-H , and H-O-H bond angles (in degrees) in methane, ammonia and water are respectively, closest to

A

109.5, 104.5, 107.1

B

109.5, 107.1, 104.5

C

104.5, 107.1, 104.5

D

107.1, 104.5, 109.5

The H-C-H , H-N-H , and H-O-H bond angles (in degrees) in methane, ammonia and water are respectively, closest to

A

109.5, 104.5, 107.1

B

109.5, 107.1, 104.5

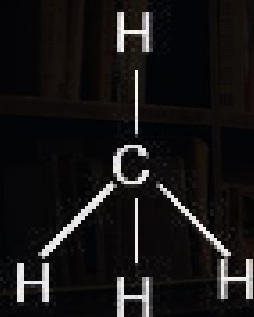
C

104.5, 107.1, 104.5

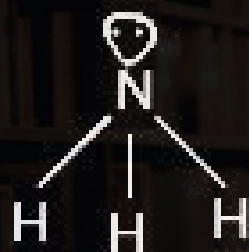
D

107.1, 104.5, 109.5

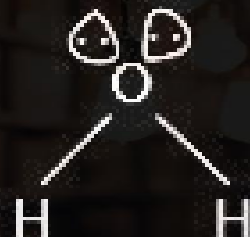
Solution:



Tetrahedral



Pyramidal
(one lone pair)



bent
(2 lone pairs)

The shape of SCl_4 is best described as a

A

square

B

tetrahedron

C

square pyramid

D

see - saw

The shape of SCl_4 is best described as a

A

square

B

tetrahedron

C

square pyramid

D

see - saw

The molecule with the highest dipole moment among the following is

A



B



C



D



The molecule with the highest dipole moment among the following is

A



B



C



D



The molecule which does not exhibit strong hydrogen bonding is :

A methyl amine

B acetic acid

C diethyl ether

D glucose

The molecule which does not exhibit strong hydrogen bonding is :

A methyl amine

B acetic acid

C diethyl ether

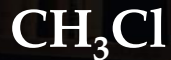
D glucose

Solution:

For H-bonding in molecule highly electronegative element & H should be directly connected. In $(\text{C}_2\text{H}_5)_2\text{O}$, H is connected to carbon.

Which of the following molecules has no dipole moment ?

A



B



C

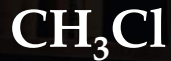


D



Which of the following molecules has no dipole moment ?

A



B



C



D



Solution:

CCl_4 has zero dipole moment due to its tetrahedral shape, all C-Cl bond moment cancel each other.

Among the following, the species with the highest bond order is :

A



B



C



D



Among the following, the species with the highest bond order is :

A



B



C



D



Solution:

According to MOT bond order of O_2 , F_2 , O_2^+ and F_2^- is respectively 2, 1, 2.5 and 0.5.

The diamagnetic species is

A

N
 O

B

NO_2

C

O_2

D

CO_2

The diamagnetic species is

A

N
 O

B

NO_2

C

O_2

D

CO_2

Contact me!

Email me at nidhi.prashar@vedantu.com

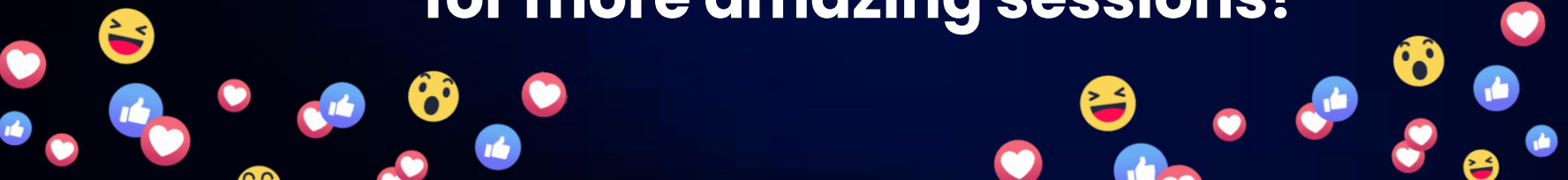
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